NATIONAL ACADEMIES Sciences Engineering Medicine

NATIONAL ACADEMIES PRESS Washington, DC

This PDF is available at https://nap.nationalacademies.org/25573



Simulation Options for Airport Planning (2019)

BUY THIS BOOK

FIND RELATED TITLES

DETAILS

78 pages | 8.5 x 11 | PAPERBACK

ISBN 978-0-309-48066-6 | DOI 10.17226/25573

CONTRIBUTORS

Florian B. Hafner, Cignus Consulting, LLC; Airport Cooperative Research Program; Transportation Research Board; National Academies of Sciences, Engineering, and Medicine

SUGGESTED CITATION

National Academies of Sciences, Engineering, and Medicine. 2019. *Simulation Options for Airport Planning*. Washington, DC: The National Academies Press. https://doi.org/10.17226/25573.

Visit the National Academies Press at nap.edu and login or register to get:

– Access to free PDF downloads of thousands of publications
– 10% off the price of print publications
– Email or social media notifications of new titles related to your interests
– Special offers and discounts

All downloadable National Academies titles are free to be used for personal and/or non-commercial academic use. Users may also freely post links to our titles on this website; non-commercial academic users are encouraged to link to the version on this website rather than distribute a downloaded PDF to ensure that all users are accessing the latest authoritative version of the work. All other uses require written permission. (Request Permission)

This PDF is protected by copyright and owned by the National Academy of Sciences; unless otherwise indicated, the National Academy of Sciences retains copyright to all materials in this PDF with all rights reserved.

AIRPORT COOPERATIVE RESEARCH PROGRAM

ACRP SYNTHESIS 98

Simulation Options for Airport Planning

A Synthesis of Airport Practice

Florian B. Hafner
Cignus Consulting, LLC
Leesburg, VA

Subscriber Categories
Aviation • Administration and Management • Planning and Forecasting

Research sponsored by the Federal Aviation Administration

The National Academies of
SCIENCES • ENGINEERING • MEDICINE
TRB
TRANSPORTATION RESEARCH BOARD
2019

Copyright National Academy of Sciences. All rights reserved.

AIRPORT COOPERATIVE RESEARCH PROGRAM

Airports are vital national resources. They serve a key role in transportation of people and goods and in regional, national, and international commerce. They are where the nation's aviation system connects with other modes of transportation and where federal responsibility for managing and regulating air traffic operations intersects with the role of state and local governments that own and operate most airports. Research is necessary to solve common operating problems, to adapt appropriate new technologies from other industries, and to introduce innovations into the airport industry. The Airport Cooperative Research Program (ACRP) serves as one of the principal means by which the airport industry can develop innovative near-term solutions to meet demands placed on it.

The need for ACRP was identified in *TRB Special Report 272: Airport Research Needs: Cooperative Solutions* in 2003, based on a study sponsored by the Federal Aviation Administration (FAA). ACRP carries out applied research on problems that are shared by airport operating agencies and not being adequately addressed by existing federal research programs. ACRP is modeled after the successful National Cooperative Highway Research Program (NCHRP) and Transit Cooperative Research Program (TCRP). ACRP undertakes research and other technical activities in various airport subject areas, including design, construction, legal, maintenance, operations, safety, policy, planning, human resources, and administration. ACRP provides a forum where airport operators can cooperatively address common operational problems.

ACRP was authorized in December 2003 as part of the Vision 100—Century of Aviation Reauthorization Act. The primary participants in the ACRP are (1) an independent governing board, the ACRP Oversight Committee (AOC), appointed by the Secretary of the U.S. Department of Transportation with representation from airport operating agencies, other stakeholders, and relevant industry organizations such as the Airports Council International-North America (ACI-NA), the American Association of Airport Executives (AAAE), the National Association of State Aviation Officials (NASAO), Airlines for America (A4A), and the Airport Consultants Council (ACC) as vital links to the airport community; (2) TRB as program manager and secretariat for the governing board; and (3) the FAA as program sponsor. In October 2005, the FAA executed a contract with the National Academy of Sciences formally initiating the program.

ACRP benefits from the cooperation and participation of airport professionals, air carriers, shippers, state and local government officials, equipment and service suppliers, other airport users, and research organizations. Each of these participants has different interests and responsibilities, and each is an integral part of this cooperative research effort.

Research problem statements for ACRP are solicited periodically but may be submitted to TRB by anyone at any time. It is the responsibility of the AOC to formulate the research program by identifying the highest priority projects and defining funding levels and expected products.

Once selected, each ACRP project is assigned to an expert panel appointed by TRB. Panels include experienced practitioners and research specialists; heavy emphasis is placed on including airport professionals, the intended users of the research products. The panels prepare project statements (requests for proposals), select contractors, and provide technical guidance and counsel throughout the life of the project. The process for developing research problem statements and selecting research agencies has been used by TRB in managing cooperative research programs since 1962. As in other TRB activities, ACRP project panels serve voluntarily without compensation.

Primary emphasis is placed on disseminating ACRP results to the intended users of the research: airport operating agencies, service providers, and academic institutions. ACRP produces a series of research reports for use by airport operators, local agencies, the FAA, and other interested parties; industry associations may arrange for workshops, training aids, field visits, webinars, and other activities to ensure that results are implemented by airport industry practitioners.

ACRP SYNTHESIS 98

Project 11-03, Topic S03-14
ISSN 1935-9187
ISBN 978-0-309-48066-6
Library of Congress Control Number 2019947480

© 2019 National Academy of Sciences. All rights reserved.

COPYRIGHT INFORMATION

Authors herein are responsible for the authenticity of their materials and for obtaining written permissions from publishers or persons who own the copyright to any previously published or copyrighted material used herein.

Cooperative Research Programs (CRP) grants permission to reproduce material in this publication for classroom and not-for-profit purposes. Permission is given with the understanding that none of the material will be used to imply TRB, AASHTO, FAA, FHWA, FMCSA, FRA, FTA, Office of the Assistant Secretary for Research and Technology, PHMSA, or TDC endorsement of a particular product, method, or practice. It is expected that those reproducing the material in this document for educational and not-for-profit uses will give appropriate acknowledgment of the source of any reprinted or reproduced material. For other uses of the material, request permission from CRP.

Cover photo: AirTOp integrated airfield, passenger terminal, and ground vehicle simulation.
Cover photo credit: Courtesy of Airtopsoft.

NOTICE

The report was reviewed by the technical panel and accepted for publication according to procedures established and overseen by the Transportation Research Board and approved by the National Academies of Sciences, Engineering, and Medicine.

The opinions and conclusions expressed or implied in this report are those of the researchers who performed the research and are not necessarily those of the Transportation Research Board; the National Academies of Sciences, Engineering, and Medicine; or the program sponsors.

The Transportation Research Board; the National Academies of Sciences, Engineering, and Medicine; and the sponsors of the Airport Cooperative Research Program do not endorse products or manufacturers. Trade or manufacturers' names appear herein solely because they are considered essential to the object of the report.

Published reports of the

AIRPORT COOPERATIVE RESEARCH PROGRAM

are available from

Transportation Research Board
Business Office
500 Fifth Street, NW
Washington, DC 20001

and can be ordered through the Internet by going to

http://www.national-academies.org

and then searching for TRB

Printed in the United States of America

Copyright National Academy of Sciences. All rights reserved.

The National Academies of
SCIENCES • ENGINEERING • MEDICINE

The **National Academy of Sciences** was established in 1863 by an Act of Congress, signed by President Lincoln, as a private, non-governmental institution to advise the nation on issues related to science and technology. Members are elected by their peers for outstanding contributions to research. Dr. Marcia McNutt is president.

The **National Academy of Engineering** was established in 1964 under the charter of the National Academy of Sciences to bring the practices of engineering to advising the nation. Members are elected by their peers for extraordinary contributions to engineering. Dr. John L. Anderson is president.

The **National Academy of Medicine** (formerly the Institute of Medicine) was established in 1970 under the charter of the National Academy of Sciences to advise the nation on medical and health issues. Members are elected by their peers for distinguished contributions to medicine and health. Dr. Victor J. Dzau is president.

The three Academies work together as the **National Academies of Sciences, Engineering, and Medicine** to provide independent, objective analysis and advice to the nation and conduct other activities to solve complex problems and inform public policy decisions. The National Academies also encourage education and research, recognize outstanding contributions to knowledge, and increase public understanding in matters of science, engineering, and medicine.

Learn more about the National Academies of Sciences, Engineering, and Medicine at **www.national-academies.org**.

The **Transportation Research Board** is one of seven major programs of the National Academies of Sciences, Engineering, and Medicine. The mission of the Transportation Research Board is to increase the benefits that transportation contributes to society by providing leadership in transportation innovation and progress through research and information exchange, conducted within a setting that is objective, interdisciplinary, and multimodal. The Board's varied committees, task forces, and panels annually engage about 7,000 engineers, scientists, and other transportation researchers and practitioners from the public and private sectors and academia, all of whom contribute their expertise in the public interest. The program is supported by state transportation departments, federal agencies including the component administrations of the U.S. Department of Transportation, and other organizations and individuals interested in the development of transportation.

Learn more about the Transportation Research Board at **www.TRB.org**.

Copyright National Academy of Sciences. All rights reserved.

COOPERATIVE RESEARCH PROGRAMS

CRP STAFF FOR ACRP SYNTHESIS 98

Christopher J. Hedges, *Director, Cooperative Research Programs*
Lori L. Sundstrom, *Deputy Director, Cooperative Research Programs*
Marci A. Greenberger, *Manager, Airport Cooperative Research Program*
Thomas J. Helms, Jr., *Senior Program Officer*
Stephanie L. Campbell, *Senior Program Assistant*
Eileen P. Delaney, *Director of Publications*
Natalie Barnes, *Associate Director of Publications*

ACRP PROJECT 11-03 PANEL

Joshua D. Abramson, *Easterwood Airport Management, College Station, TX* (Chair)
Debbie K. Alke, *Montana DOT, Helena, MT* (Retired)
Gloria G. Bender, *TransSolutions, LLC, Fort Worth, TX*
David A. Byers, *Quadrex Aviation, LLC, Melbourne, FL*
David N. Edwards, Jr., *Greenville–Spartanburg Airport District, Greer, SC*
Brenda L. Enos, *Burns & McDonnell, Kansas City, MO*
Linda Howard, *Independent Aviation Consultant, Bastrop, TX*
Patrick W. Magnotta, *FAA Liaison*
Matthew J. Griffin, *Airport Consultants Council Liaison*
Liying Gu, *Airports Council International—North America Liaison*
Adam Williams, *Aircraft Owners & Pilots Association Liaison*
Christine Gerencher, *TRB Liaison*

TOPIC S03-14 PANEL

Jennifer Dermody, *Metropolitan Washington Airports Authority, Washington, DC*
Michael D. Floyd, *JACOBS Global Buildings, Advance Planning Group, Atlanta, GA*
Eugene Gilbo, *Volpe Center, Chestnut Hill, MA* (Retired)
Brent Kelley, *Corgan, Los Angeles, CA*
Majed Khater, *Clark County (NV) Department of Aviation, Las Vegas, NV*
Rongfang "Rachel" Liu, *New Jersey Institute of Technology, Newark, NJ*
Erik Wilkins, *Ricondo & Associates, Inc., Chicago, IL*
William Reinhardt, *FAA Liaison*
Christopher J. Oswald, *Airports Council International—North America Liaison*
Christine Gerencher, *TRB Liaison*

Copyright National Academy of Sciences. All rights reserved.

ABOUT THE ACRP SYNTHESIS PROGRAM

Airport administrators, engineers, and researchers often face problems for which information already exists, either in documented form or as undocumented experience and practice. This information may be fragmented, scattered, and unevaluated. As a consequence, full knowledge of what has been learned about a problem may not be brought to bear on its solution. Costly research findings may go unused, valuable experience may be overlooked, and due consideration may not be given to recommended practices for solving or alleviating the problem.

There is information on nearly every subject of concern to the airport industry. Much of it derives from research or from the work of practitioners faced with problems in their day-to-day work. To provide a systematic means for assembling and evaluating such useful information and to make it available to the entire airport community, the Airport Cooperative Research Program authorized the Transportation Research Board to undertake a continuing project. This project, ACRP Project 11-03, "Synthesis of Information Related to Airport Practices," searches out and synthesizes useful knowledge from all available sources and prepares concise, documented reports on specific topics. Reports from this endeavor constitute an ACRP report series, Synthesis of Airport Practice.

This synthesis series reports on current knowledge and practice, in a compact format, without the detailed directions usually found in handbooks or design manuals. Each report in the series provides a compendium of the best knowledge available on those measures found to be the most successful in resolving specific problems.

FOREWORD

By Thomas J. Helms, Jr.
Staff Officer
Transportation Research Board

Airports are dynamic environments in which multiple modes of transportation can connect to efficiently convey people and goods throughout the United States and around the world. Simulation tools can play an important role in assisting airport administrators, designers, and engineers with planning for any physical and operational changes to an airport. In some cases, the completion of simulation studies is a de facto requirement when planning airport development projects.

One of the challenges that airports face in using simulation to support planning activities is selecting the tool that best serves the information needs of the project. Thus, the focus of this report is on available simulation capabilities and how airports employ those capabilities.

This study is based on information acquired through a literature review and the results of a survey intended to elicit responses about the typical and specific application of simulation tools by airports. *ACRP Synthesis 98* presents the different simulation studies common in the airport planning context, the typical inputs and outputs for the different simulation studies, and information about the various simulation tools available to airports. The report also includes some case examples of airports using simulation for different planning purposes.

Florian B. Hafner of Cignus Consultants, LLC, in Leesburg, Virginia, synthesized the information and wrote the report. The members of the topic panel are acknowledged on Page iv. This synthesis is an immediately useful document that records the practices that were acceptable within the limitations of the knowledge available at the time of its preparation. As progress in research and practice continues, new knowledge will be added to that now at hand.

Copyright National Academy of Sciences. All rights reserved.

Copyright National Academy of Sciences. All rights reserved.

CONTENTS

1 Summary

3 **Chapter 1** Introduction
3 Background
5 Study Approach

6 **Chapter 2** Basic Simulation Concepts for Airport Planning and Design
7 Modeling Versus Simulation
8 Simulation Process
9 Simulation Calibration and Validation
10 Simulation Data Requirements
11 Analysis and Metrics
12 Quality Assurance and Control

14 **Chapter 3** Industry Perspectives of Simulation Tools for Airport Planning and Design
14 Airspace
16 Airport or Airfield
19 Terminal Passenger Flow
21 Terminal Systems
22 Ground-Access Simulation
23 Simulation Results and Application
24 Simulation Model Maintenance

25 **Chapter 4** Simulation Tool Application Guidelines
26 Master Planning
29 Construction Phasing Plans and Impact Studies
30 Airport Capacity and Delay Studies
32 Terminal Passenger Flows
34 Terminal System Design and Planning
35 Ground-Access Studies
36 Airspace Procedure Design
37 Air Traffic Management Concept Evaluation
38 Metrics and Analysis

41 **Chapter 5** Simulation Tool Characteristics and Capabilities
41 Airspace Analysis and Simulation Tools
41 Airfield and Airport Analysis and Simulation Tools
42 Terminal Passenger Flow Analysis and Simulation Tools
42 Airport Terminal Systems Analysis and Simulation Tools
42 Curbside Vehicle Analysis and Simulation Tools

Copyright National Academy of Sciences. All rights reserved.

45 **Chapter 6** Conclusions and Future Research
45 Findings
46 Capability Gaps and Problem Areas
47 Future Research

48 References

50 Abbreviations and Acronyms

51 **Appendix A** Survey Questionnaire

60 **Appendix B** Sample Simulation Study Inputs and Outputs

62 **Appendix C** Simulation Model Characteristics

Note: Photographs, figures, and tables in this report may have been converted from color to grayscale for printing. The electronic version of the report (posted on the web at www.trb.org) retains the color versions.

Copyright National Academy of Sciences. All rights reserved.

SUMMARY

Simulation Options for Airport Planning

Airports are transportation hubs where several modes come together to transport passengers, baggage, and cargo. Many airports also provide commercial outlets for products and services. In such a dynamic and interconnected environment, simulation plays an integral role in the planning of operational and physical changes. Credible simulation projects can help airport administrators, designers, engineers, and planners estimate the impact of planned changes on passenger traffic, aircraft traffic, roadway traffic, baggage movements, and other subsystems such as bus and train links and aircraft ground support operations.

To that end, the report from Airport Cooperative Research Program Project 11-03/ Topic S03-14 is the result of a study that focused on collecting and analyzing information on current industry practices and on applications of simulation tools for airport planning and design.

The primary objective of this report is to support decision makers and practitioners with their selection of simulation tools based on their specific environment and project types. The report attempts to answer the following questions for airport planners as they endeavor to select the best and most appropriate tool:

- Which simulation tools are available for my type of project?
- Which simulation tools and methods have other airports of similar size, fleet mix, and other operating characteristics used for similar projects?
- How would an airport planner choose a simulation tool for a specific project?
- What level of detail is appropriate based on the project scope, resource availability, and information requirements?
- What simulation metrics and visualization capabilities are required to support the project goals and decision-making process?

The information collected for this study is focused on the application and current usage of fast-time simulation tools by various airports, airport planners, design/build companies, and FAA. Selected case examples are provided to demonstrate the use of simulation tools for the most common project types and areas. The report presents a detailed database of simulation tools, capabilities, and basic usage requirements. All of this information feeds a decision framework for airport planners and designers to select the most appropriate tool.

In addition to a detailed literature review of current industry practices and case examples, this report uses an industry survey of representative airport planners, consultants, and government organizations to gain more insight into specific simulation projects and tools as applied in airport planning and design. Using published case examples and the results from the survey, the information on simulation tools is categorized by area of applicability—airspace, airside, terminal, curbside—and subsequently by the type of project. Based on the

Copyright National Academy of Sciences. All rights reserved.

survey results, the project type—for example, master plan or capacity and delay study—appeared to be the driving factor in the selection of simulation tools. For each project type, the report lists different simulation tools that can be used, their basic characteristics, and their capabilities.

In addition to guidelines for applying simulation tools based on current industry trends and practices, the report provides detailed information about specific simulation tools used by airport planners and designers today. Combined with selection guidelines, this detailed database of simulation tools can further guide simulation tool selections based on other characteristics such as major functionalities, data and resource requirements, and analysis and output capabilities.

The information presented in this report comprises recent published literature and various other sources, including

- Airport planner and designer industry survey.
- Simulation tool developer literature and online resources.
- Recent use cases and project descriptions.

Statistical information collected from the survey and recent case study information directly fed the simulation tool selection framework. Additional information on simulation tools is provided as a general reference, but this information can also support the selection process.

As simulation technologies and applications are ever-evolving, the information represents a snapshot of current practices and simulation tools at the time this report was published. In addition to the report, there are three appendices. Appendix A is the survey questionnaire, Appendix B describes sample simulation study inputs and outputs, and Appendix C describes simulation model characteristics.

Copyright National Academy of Sciences. All rights reserved.

CHAPTER 1

Introduction

Global business and tourism depend heavily on the efficient operation of airports and movement of passengers, baggage, and cargo across many areas. With increasing demand and connectivity requirements for airports comes the need for more sophisticated simulation and modeling tools to validate design assumptions. Furthermore, airport design and planning decisions have significant impacts on policy and major capital improvement decisions, which can be supported by simulation and modeling tools at many levels.

Background

Airports form an integral part of the nation's transportation system. They represent the connection of multiple transportation modes where passengers transition between ground transportation (e.g., automobiles, trains, and buses), passenger terminals, and aircraft. They are highly complex and dynamic systems that require the use of detailed planning and design processes to ensure efficiency, security, and profitability.

An airport is much more than the sum of its parts. Modern airports represent a complex network of interconnected and dependent systems where every subsystem must operate and work together to maximize the efficiency and functionality of the overall airport operation. The airport system is composed of several components—including airspace, airfield, terminal, and curbside operations—all of which require strategic and day-to-day operational planning.

In this dynamic environment, design plans and operations constantly evolve. Design decisions impact strategic capital improvements and changes in operational demand also require tactical changes in airport operations planning.

Many airports and airport planners around the nation use simulation tools to support near-term tactical decision making as well as strategic planning projects. The toolsets and processes used to analyze and simulate airport operations have changed significantly since the 1980s, when analysis techniques were limited to general purpose queuing and network analysis concepts or purpose-built simulation tools (Gosling, 1988). These tools have become much more sophisticated and accurate in emulating real-world aircraft, passenger, and vehicle dynamics. Current simulation tools can model detailed operational scenarios and conduct analyses of highly complex airport processes, including airspace and airfield operations, terminal passenger traffic, baggage systems, curbside vehicle traffic, and many other subprocesses. Virtually all major airport development efforts today include detailed airport simulation efforts to support airspace procedure designs, master plans, capacity-demand studies, terminal passenger flow plans, curbside capacity studies, and environmental impact assessments, among others.

Simulation tool capabilities range from simple stochastic analyses to more detailed queueing models to very complex processes and interactions. As simulation tools become more

Copyright National Academy of Sciences. All rights reserved.

sophisticated and powerful, they are capable of modeling increasingly realistic aircraft, passenger, and vehicle operations at highly detailed levels. Current-day simulations are capable of very accurately predicting future operational conditions, and the survey results indicate a strong desire to use simulation tools across a variety of organizations and for a variety of projects.

Evolving simulation tool complexity—mainly in terms of functionality, data processing requirements, and results analysis—comes at the price of increased training requirements, resources, and licensing fees. Whether a study is focused on assessing changes in airport infrastructure (runways, taxiways, gates, and terminals), aircraft and passenger traffic patterns, baggage system configurations, or environmental or noise patterns, the choice of simulation tool can have a significant impact on the credibility of the study, as well as on the study budget and schedule.

Generally, simulation tools can be categorized by their fidelity and capabilities, as illustrated in Figure 1. For this discussion, the term "fidelity" refers to the faithfulness of the technological and simulation tool behavior in representing real-world processes and objects. A high-fidelity, or microscopic, tool can generally simulate objects and interactions with high detail and precision. A low-fidelity, macroscopic simulation tool, on the other hand, usually uses abstractions and summary data for analyses and predictions.

Analyses can range from table lookups to medium-fidelity analytical and mathematical models to very high-fidelity capacity and delay simulation applications. Each of these types of analysis tools serves specific purposes and should be applied to answer specific questions and support airport planning analyses.

However, the applicability and usefulness of specific simulation tools may not be widely understood by all airport planners and designers. Many times, simulation tool decisions are made based on resource availability and other factors without an appropriate assessment of needs, data availability, and metric and reporting requirements. Sometimes it may even become necessary to use multiple tools in combination (e.g., terminal passenger models connected to airside aircraft models), so interfaces and data input and output characteristics need to be understood.

Several reports and articles describe current practices and case studies of applying simulation tools to assess airport, airfield, and terminal capacity and delays (LeighFisher et al., 2012).

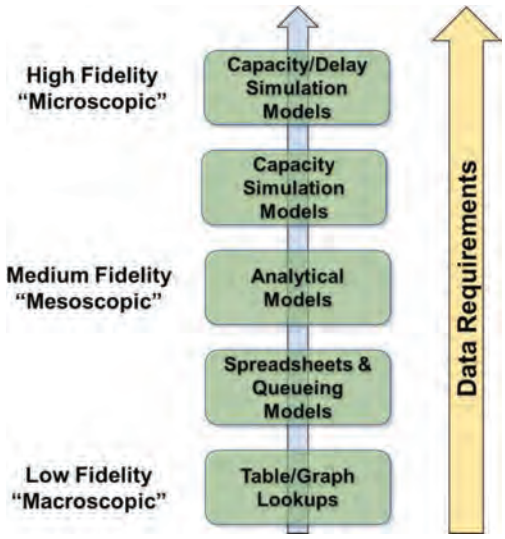


Figure 1. Simulation model fidelity.

Copyright National Academy of Sciences. All rights reserved.

However, the number and types of simulation tools available to airport planners can still cause confusion and inappropriate application of a simulation.

Study Approach

This synthesis report explores the functional and analytical capabilities of simulation tools used by the nation's airport planners to simulate and analyze airport or airfield, airspace, terminal, and ground-access operations. It focuses not only on simulation software capabilities, but also on the industry application and best practices when using these tools on representative projects.

With a focus on how simulation tools are used at medium and large airports, a review of recent literature and project reports forms the baseline of the current environment description. This review is used to provide airport planners with background on individual simulation tools and potential case examples. It is supplemented by an industry survey of airport planners and designers that was used to capture current practice trends and case examples from recent airport planning on representative projects. The survey that was conducted for this report was designed to elicit simulation-specific practices and opinions and was specifically targeted at airport organizations and planners that may currently be using—or have used in the past—simulation tools to support airport planning. The survey results are included in various parts of this report to demonstrate simulation tool usage and provide ideas and relevant information to readers. A complete list of all survey questions can be found in Appendix A.

The literature review and survey information collected are used to build a framework for guiding airport planners that are new to simulation tools and their applicability in the most appropriate tool selection based on their environment and project needs. The report proposes simulation tool selection strategies based on significant project and organizational characteristics. Using this strategy as well as case examples, readers can use their own project needs and draw parallels with current industry simulation tool application trends. In addition, more detailed information on simulation tool characteristics and capabilities can serve as further reference for the airport planner in this selection process. In some cases, direct communication with simulation tool developers and information from online websites and resources is also used to support documented capabilities and identify capability gaps for specific simulation software tools.

Copyright National Academy of Sciences. All rights reserved.

CHAPTER 2

Basic Simulation Concepts for Airport Planning and Design

At the beginning of a simulation study, analysts generally ask themselves certain questions that can help them frame the problem scope and determine analyses concepts. The types of questions that are generally asked include:

- Which parts of the airfield, airspace, terminal structure, and roadways should be simulated?
- What simulation tool fidelity is appropriate for the analysis, available resources, and overall project scope?
- What input data are available and needed to build the models?
- Which metrics will support the larger analyses and are meaningful?
- What visualization and animation capabilities are desired for analysis and presentation?
- Which tools and methods have other organizations used for similar projects and environments?

The answers to these types of questions drive simulation tool choices. They were also used to guide the information contained in this report in terms of the overall selection framework and relevant information presented for each simulation tool.

Simulation studies can vary significantly in terms of complexity and resource and data requirements, but the knowledge of a specific simulation tool needs to be combined with a fundamental understanding of simulation and analysis concepts to produce meaningful results. In addition to the ability to use a specific software or concept, the execution of a successful simulation and analysis project depends heavily on the understanding of mathematical, analytical, and fast-time simulation concepts. This section of the report provides basic information on simulation processes, data gathering, and data requirements, as well as commonly used analysis metrics for the reader. Understanding some of these basic concepts is a key factor in choosing simulation tools for specific projects.

With increased simulation tool complexity and functionality capabilities comes the need for more informed and structured simulation development and analysis methods. Whereas basic concepts of capacity, delay, and queueing theory may be sufficient to conduct macroscopic level studies, high-fidelity microscopic simulation tools need an increased level of expertise, resources, and input data. Their ability to model detailed objects as well as to incorporate dynamic demand patterns requires a detailed understanding of simulation methodologies, input and output data, metrics, and analysis techniques. This is particularly true in an increasingly connected world where simulation technology is used to address large-scale system-of-system problems at a complex level with competing objectives (Nelson, 2016).

Regardless of the level of detail of analysis or how outputs from the simulation analysis are intended to be used, the methodologies need to be sound and the results defendable. This is particularly true considering the size of capital investment decisions that are sometimes supported by the simulation results.

Copyright National Academy of Sciences. All rights reserved.

Moreover, the use of simulation tools should be integrated into a collaborative process that involves all affected stakeholders. At every step in the process, educated and collaborative decisions need to be made, starting with the level of analysis required, data availability and requirements, resource availability, simulation tool selection, calibration capabilities and requirements, design of experiments, and understanding of the limitations of the simulation tools themselves.

Modeling Versus Simulation

One of the most common misunderstandings about simulation studies is the difference between modeling and simulation. While many practitioners use these terms interchangeably and in sequence, they technically have different meanings. These differences should be understood and communicated to minimize misunderstandings across stakeholders and within the simulation study team.

Models are representations of reality or concepts that can be useful for visualization purposes or to perform "static" analyses. This may include three-dimensional (3-D) or even four-dimensional (4-D) visualizations of aircraft or passenger operations.

Simulations, on the other hand, provide a "dynamic" capability that allows the user to change system and environment variables and observe the perceived differences in operation. So, the purpose of simulation is to study system behavior through the manipulation of variables, that is, the addition of runways, changes in terminal layouts, etc., that cannot be, or are impractical to be, changed in real life.

In essence, the model can be considered a product or an instance of an operation. A simulation is the process of using models for further analysis and decision-support purposes. Understanding this difference can be crucial in the setting of expectations and the understanding of why and when simulation tools are used.

Throughout a planning study, both static models and dynamic simulations may be used in the decision-making process. Typically, more static analyses using spreadsheet models and tables are used in the initial stages of a project to assess high-level capacity and delay constraints. Results from this initial analysis then justify and feed higher-fidelity dynamic simulation studies. Although this needs to be evaluated on a project-by-project basis, certain types of projects suggest the use of simulation tools in addition to static models. These include:

- Projects that focus on "what-if" studies to determine the best possible solution and that involve significant levels of uncertainty.
- Projects of high cost or visibility.
- Projects in which dynamic visualizations of objects are important.
- Projects in which the likely outcomes result in environmental studies, regulatory changes, or other significant impacts on stakeholders.
- Projects with multiple focus areas (i.e., airfield, ground, terminal) and multiple potential bottlenecks.
- Projects for which it is infeasible or impractical to gather all operational data to construct static prediction models.
- Projects for which mathematical simplifications are not feasible.

Another important aspect of simulation studies that should also not be misconstrued is simulation-based optimization. Some airport planners consider simulation tools as a means for optimizing airside operations, passenger flows, and resource use. In fact, very few simulation tools—mainly the more generic tools—offer optimization modules where inputs are varied

Copyright National Academy of Sciences. All rights reserved.

structurally until a specific objective is maximized. The vast majority of simulation tools simply run scenarios and environments that have been defined and developed by the simulation analyst with very limited sensitivity analyses based on specific randomized input variables. It is up to the simulation analyst to develop an experimental plan that explores alternative what-if scenarios sufficiently to be able to determine the "best" solution for a specific study. This by no means guarantees an optimal solution.

Simulation Process

An analysis of airport, passenger, or curbside operations can take many forms and be accomplished using various analytical and simulation tools. From simple spreadsheet models to queueing theory analyses to highly detailed dynamic simulation tools, the general concepts and tools that support these studies include:

- *Spreadsheet Models* based on prediction models or processes using historically observed data.
- *Queueing Theory,* in which dynamic demand can be evaluated on a network of capacitated resources such as check-in counters and security check lines and overall delays and resource utilization are captured.
- *Optimization Techniques* that use such concepts as dynamic programming to estimate maximum throughputs, resource utilization, etc.
- *Monte Carlo Simulation*, which uses repeated random sampling based on historical probabilistic distributions of such processes as security check times and passenger transit times to obtain statistically valid and more realistic results for the evaluation of operational processes.
- *Discrete-Event Simulation,* which is the process of structuring and modeling the behavior of a complex system as a discrete sequence of events in time where object states and environmental variables change over time.

All of these different concepts support airport planning and design decisions at specific levels and for very specific purposes. Some lend themselves more toward airfield or airspace studies, whereas others are really developed for passenger and process flows. All tools produce a plethora of output data and require the modelers to interpret, process, and display information. Some tools provide better model visualization capabilities than others.

Airport planners today use different fidelity simulation tools to support decision making at various levels through the use and evaluation of what-if alternative scenarios. A basic simulation study process is illustrated in Figure 2. Although this process is logically laid out to be linear and sequential, most simulation studies will iterate through various phases multiple times to account for errors in data and interpretations and changes in requirements and to manage internal simulation variability that detracts from the study methodology and mission.

An important aspect of simulation study project management is the development of a sound execution plan. This can be particularly valuable for simulation studies with large numbers of scenarios. The *experimental design* defines basic information about each simulation scenario and can be a valuable tool to depict simulation input changes in what-if scenarios and guide subsequent analyses. It can be used to demonstrate to team members which scenario variables are fixed and which are changing and thus subject to analysis.

Another important factor to consider for simulation studies is the idea of *multiruns*, or the incorporation of statistical uncertainty into otherwise deterministic simulations. This practice has its roots in Monte Carlo simulation and adds a level of statistical validity to the output metrics based on random natural variations to input parameters such as demand profiles, entity characteristics or behavior, and resource processing times.

Copyright National Academy of Sciences. All rights reserved.

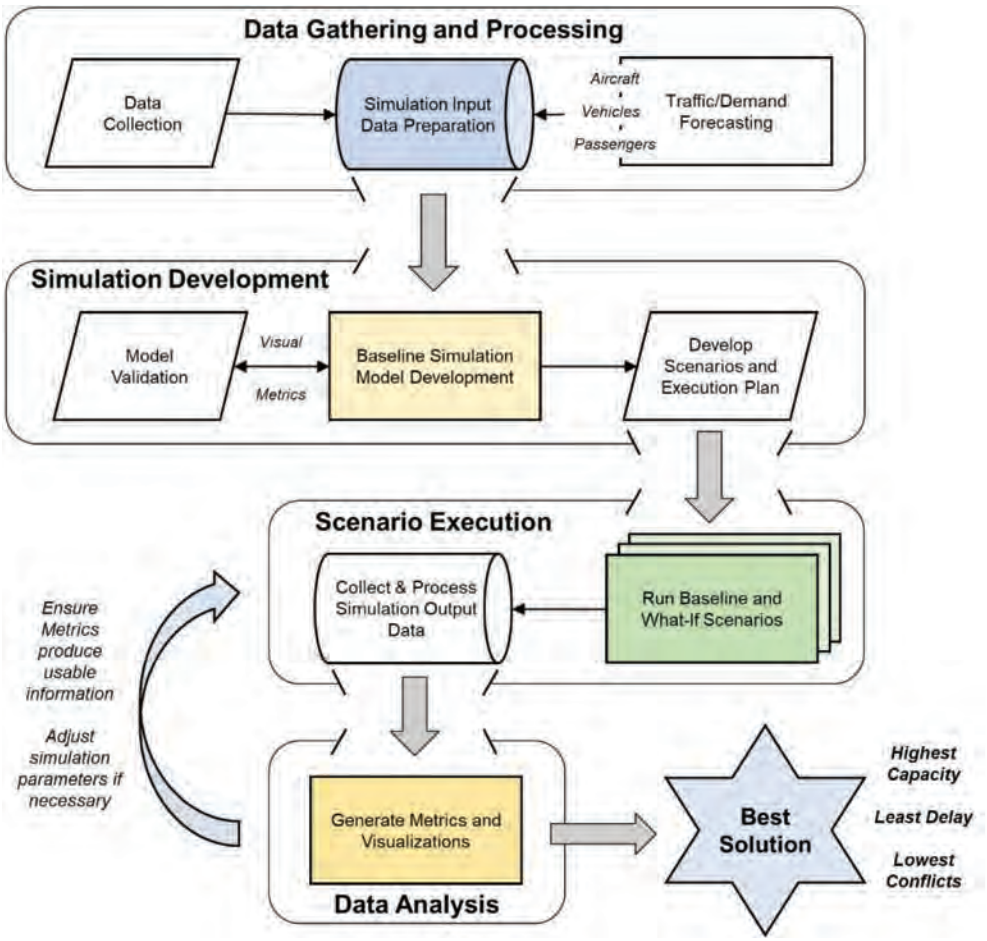


Figure 2. Basic simulation study process.

Simulation Calibration and Validation

Before any what-if scenarios are tested and simulated, however, the baseline simulation models should be calibrated and validated to ensure that they reflect current reality, or intended reality, within reason. This can be accomplished visually for high-fidelity tools, as well as by using simulation output metrics. For instance, simulation models can be calibrated against historical busy-day observed throughputs and delays to ensure they represent a valid baseline for further study. Some metrics and resources that may be useful for calibrating baseline models include

- ASPM/OpsNet Data for hourly throughputs, weather conditions, and runway configurations (FAA website access may be required).
- Bureau of Transportation Statistics (BTS) data for taxi times and airline-reported delays.
- System wide information management (SWIM) Traffic Flow Management (TFM) data for aircraft tracks, throughputs, and flight schedules (FAA SWIM data access required).
- Airport Surface Detection Equipment, Model X (ASDE-X) data for ground tracks and arrival and departure records (FAA data access required).
- Airline postoperational analysis data from tools such as POET (Post Operations Evaluation Tool).

Simulation models can subsequently be validated operationally and visually by subject matter experts who are familiar with the given operation. These experts may include airport personnel

Copyright National Academy of Sciences. All rights reserved.

responsible for managing operations, air traffic controllers, or airline representatives. If simulation models are intended to be used for future studies, periodic recalibration may also be required to keep them up-to-date and ensure that infrastructure, environmental, operational, and procedural changes are reflected.

What should be understood in this calibration and validation process is that a simulation of a specific operational environment will never exactly match—or replicate—the actual operation. Rather than trying to exactly match historic operational characteristics, the analyst may be better off focusing on trend analyses and ranges that are aggregated for a specific configuration, weather pattern, or demand level. A calibration against historical data should also appreciate the variability in the system and that periods of irregular operations may skew historical data analyses. An experienced analyst will need to manage simulation project expectations and ensure that the important aspects of the models are "good enough" or reasonable from an operational reality perspective and subsequent what-if analyses.

It should also be noted that FAA does not prescribe the use of specific simulation tools for airfield, airspace, and terminal planning studies. The choice of which simulation tool to use for a study is still very much left to the simulation analyst and sponsor of the analysis. However, to consider the simulation study as valid, FAA requires that studies using new simulation tools—those that have not previously been used on similar studies for FAA—contain a "simulation tool validation" process step. This step should not only validate that baseline simulations can be calibrated against real-world data but also that the simulation tool is internally verified and that output data is validated.

Simulation Data Requirements

Input data requirements for analytical and simulation tools change according to the level of fidelity. The level of detail required also varies depending on the area of applicability, the study methodology, and the intended analysis metrics. Generally, the types of simulation input data can be categorized into:

- *Published data* such as airport and terminal layouts, procedures, flight schedules, and aircraft performance characteristics.
- *Collected data*, which include information that is derived from historical data or manually collected such as process times, arrival or departure separations, etc.
- *Visual/observed data*, which are based on the direct observation of specific processes and operational procedures.

The collected information is then transformed into simulation input data. The following datasets represent several common simulation inputs (a more complete listing can be found in Appendix B).

- Flight, passenger, and vehicle demand schedules.
- Terminal passenger characteristics and process time distributions.
- Airport or airfield, airspace, terminal, and roadway designs and configuration.
- Vehicle configurations and performance characteristics.
- Operating rules and restrictions.

As an example, a runway capacity study may simply require information on the expected busy arrival or departure demand period and fleet mix. This demand information can simply reflect a single busy design hour to evaluate scenarios under high demand or load. However, a higher fidelity dynamic airport or airside simulation will need either a flight schedule based on a full historic high season, a busy day, or a detailed design-day flight schedule (DDFS) that is specially created to provide a clean and "pure" demand scenario based on known data and variables.

Copyright National Academy of Sciences. All rights reserved.

Care should be taken, however, to ensure the DDFS accurately reflects historic demand trends (Landrum & Brown et al., 2010).

The collection, processing, and integration of data throughout the development of the simulation models also require care. A common problem that is repeated by many industry practitioners and the importance of which cannot be understated is that simulation study results and analyses are only as accurate as the data used to develop the models.

Analysis and Metrics

The effective use of simulation tools also requires analysts to have strong statistical analysis capabilities and an understanding of how the system works. A basic understanding of means, standard deviations, confidence intervals, histograms, and other basic statistics forms the basis of sound simulation analyses practices and produces results that are meaningful and defendable.

Core metrics form the key evaluation criteria for simulation scenarios. These metrics are primarily numerical and based on mathematical derivations of simulation outputs, but they may also be in the form of derived visual representations of numerical data (e.g., maps, charts, and other infographics). A good analyst will usually find ways of extracting information from simulation tool output data to support the conclusions of a study. A description of several commonly produced metrics for airport planning and design studies follows.

Throughput and Capacity

From an airport or airfield perspective, capacity is a measure of the maximum number of aircraft operations that can be accommodated on the airport or by an airport component within a given period of time.

Capacity calculations range from low-fidelity estimates based on simple assumptions about the runway layout to quick spreadsheet models to complex and high-fidelity simulation results (TransSolutions et al., 2014).

Throughput and capacity can be analyzed in various ways. Lower fidelity analytical models will generally produce single summary statistics for capacity that can describe

- Maximum capacity.
- Maximum sustainable capacity.
- Practical capacity.

At the most detailed level, graphs and tables of daily or hourly throughput—based on the DDFS or other dynamic demand tables—can be used to demonstrate daily throughput variations and provide a detailed comparison of how well a system can meet anticipated daily or hourly demand patterns. Generally, hourly throughputs are used as the basis for reports and as a metric for summary statistics of throughputs and capacity.

Delay

Delay can be associated with almost all objects on the airfield and within or near the terminal. Aircraft encounter delays in the airspace, at the departure runway, throughout taxiing, and at the gate. Passenger flow models will show delays at various queueing locations, including check-in counters, security checkpoints, and the like. Vehicles on roadways can experience congestion at various locations where they access or exit an airport.

Copyright National Academy of Sciences. All rights reserved.

Capacity-driven delays can be accurately predicted using higher detail simulation software. Generally, to calculate delays, a nominal or unimpeded time to complete a specific process of function is calculated, and then any additional time is measured as "delay" (TransSolutions et al., 2014).

However, the underlying simulation events for this delay calculation may differ slightly based on the simulation tool, and the differences need to be understood by the modeler and project sponsor. Some simulation tools may simply provide summations of individual process delays as a total delay for an aircraft, passenger, or vehicle without taking into consideration excess time experienced by longer travel paths and routine delays. For example, the Total Airspace and Airport Modeler (TAAM) does not natively count taxiway reroutes or pushback pauses in delay calculations. This can be counterintuitive to airline definitions of ground delay, which simply measure the differences in expected arrival, departure, or transit times versus actual times.

Delays can be used to support cost–benefit analysis (CBA) by simply applying cost factors to individual delay types: aircraft operating cost (air and ground), aircraft nonoperating cost, and passenger time cost. A more detailed FAA description of guidance to support capacity-driven CBA studies can be found on the FAA's regulations and policies website (FAA, 2017).

Annualization

Often delays and associated costs also need to be annualized to support CBAs. Care should be taken in the annualization of airspace and airfield simulation data if meteorological information is used as a basis for this scaling. It is common practice to use periodic airport meteorological weather information (METARs) in this analysis, but some airports are more affected by convective weather forecasts at specific times of the year, which are frequently ignored.

Simulation-based daily metrics can also be scaled using annual service volume information—either aircraft, passengers, or vehicles—as well as proportion-of-time information for each configuration modeled. However, the analyst should take care in defining a sound methodology for this approach as it can be based on various factors, including weighted capacity and throughput information or aircraft and passenger delays.

Other Metrics

Other metrics that are frequently used in the analysis of simulation data include:

- Airspace sector throughput and occupancy.
- Resource usage (gates, check-in counters, security checkpoints, curb allocation).
- Workload (primarily air traffic control [ATC]-related).

In addition to detailed analysis statistics, other metrics are frequently used to visualize and quantify areas of interest or concern. These types of secondary metrics are geared toward high-fidelity simulation tools and can include:

- Choke points where high localized delays are experienced.
- Conflict points where airspace, airfield, or passenger flows frequently cross or conflict.
- Heat maps displaying "busy" areas with high traffic volume.

Quality Assurance and Control

As with any simulation project, there are best practices and processes that should be considered to ensure that a study produces high quality as well as dependable results. This is particularly important considering the capital costs and operational and safety impacts of the

Copyright National Academy of Sciences. All rights reserved.

types of projects that airport planners typically consider for simulation studies. In addition to a thorough understanding of the simulation tools that are to be used, as well as experience in managing simulation studies, some of the basic quality assurance (QA) concepts that should be incorporated into all simulation studies include:

- The use of industry best practices for developing simulation scenarios, extracting metrics, and analyzing results. A specific example of best practices that is available to simulation analysts is the TAAM Best Practices Guideline. This document assembled a collection of industry best practices in the context of project planning and execution for applying TAAM to aviation systems simulations (Boesel et al., 2001). Similarly, Eurocontrol documented a TAAM Operational Evaluation project that also included many best practices for conducting fast-time airport and airspace simulation studies that extend beyond the use of the TAAM software (Sillard et al., 2000). Although some of these documents may be quite outdated in terms of simulation tool functionality and versions, the basic principles illustrated in these guidelines can still be very applicable.
- A focus on the development of use cases and an understanding of variables and metrics at the beginning of the project rather than as the project evolves. The early identification of simulation project risks, data availability issues, and other potential shortcomings is key in maximizing the potential of a study and minimizing simulation study execution risks.
- Basic systems engineering principles such as verification and validation of simulation scenarios against the study scope and requirements and use of a sound documentation and change control process.
- A sound information and document version control system that ensures simulation input data integrity across numerous scenarios and tracks changes in simulation input assumptions and derived input data.
- A detailed understanding of the roles and responsibilities across the simulation team, including simulation analysts, airport staff, airlines, and other aircraft operators. Seasoned simulation analysts will set and manage expectations at the beginning of a project to minimize the risk of failure.

Copyright National Academy of Sciences. All rights reserved.

CHAPTER 3

Industry Perspectives of Simulation Tools for Airport Planning and Design

Simulation tools can be used on a wide range of projects and for varying purposes. To focus the discussion for this report, information collected from the study survey is combined with relevant simulation studies found in recent industry publications to provide the reader a holistic view on simulation study project types for airport planning and design. This chapter also tries to provide an idea of the required resources, simulation model maintenance requirements, and general practices associated with specific simulation tools.

The use of simulation tools to support airport planning and design studies is very popular among airport planners and consultants and is, in some cases, required by FAA to support major airport or airspace design projects. Although FAA does not mandate the use of specific tools, the overall use of simulation tools is a de facto requirement for any major planning effort. From an airport planning and design perspective, simulation and modeling tools can support several different areas and products.

As Figure 3 shows, the use of simulation tools is very popular for airport, airspace, and terminal applications, but they are also sometimes used to analyze other transit applications. The survey results and literature review also show that the types of simulation tools used by industry seem to depend heavily on factors such as the airport operational environment and airport size. The survey also indicates that *capacity/demand planning* is likely the most popular area for use of simulation tools, with *master planning* projects also often using them.

The most common perceived benefits of conducting simulation studies in support of airport planning and design include proofs-of-concept for decision support, the generation of data to support policy and financial analyses, design decision support, and long-term capacity planning. Figure 4 illustrates these benefits and their relative importance levels based on the survey results.

Generally, analytical and simulation models can be categorized based on their area of applicability, level of fidelity, and analysis capabilities. As Figure 5 shows, more than 90 percent of survey respondents indicated that they use purpose-built commercial simulation tools for airport planning and design purposes. The same percentage of respondents also use these commercial tools for airspace studies. About 30 percent of those surveyed also noted that they support these analyses using spreadsheet or other in-house models. This shows that higher fidelity tools are also frequently supported by generic applications, including spreadsheet tables and basic queueing models that use generic software tools such as Arena or Simio.

Airspace

By far the most common simulation project for airspace studies is the assessment of airspace restrictions on runway and airport capacity. This includes simulation-based evaluations of proposed airspace procedures as well as impacts of TFM decisions on airport ground operations.

Copyright National Academy of Sciences. All rights reserved.

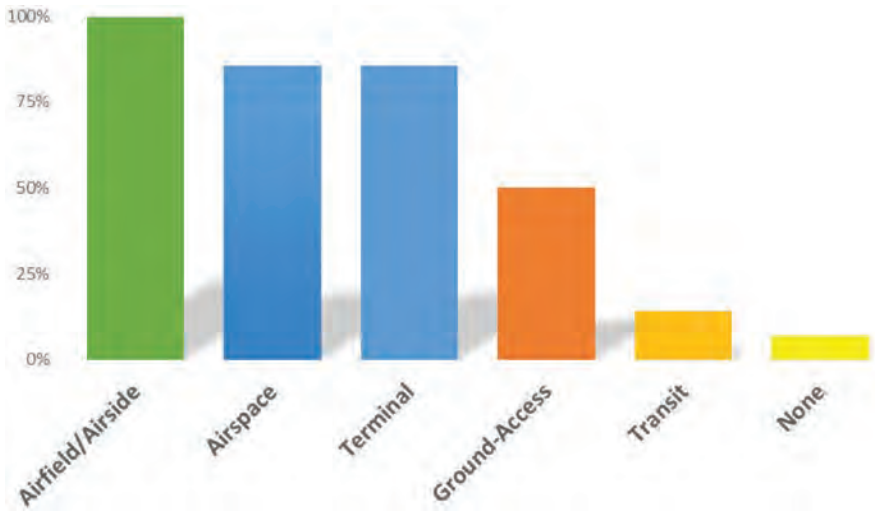


Figure 3. Survey results—simulation use by area.

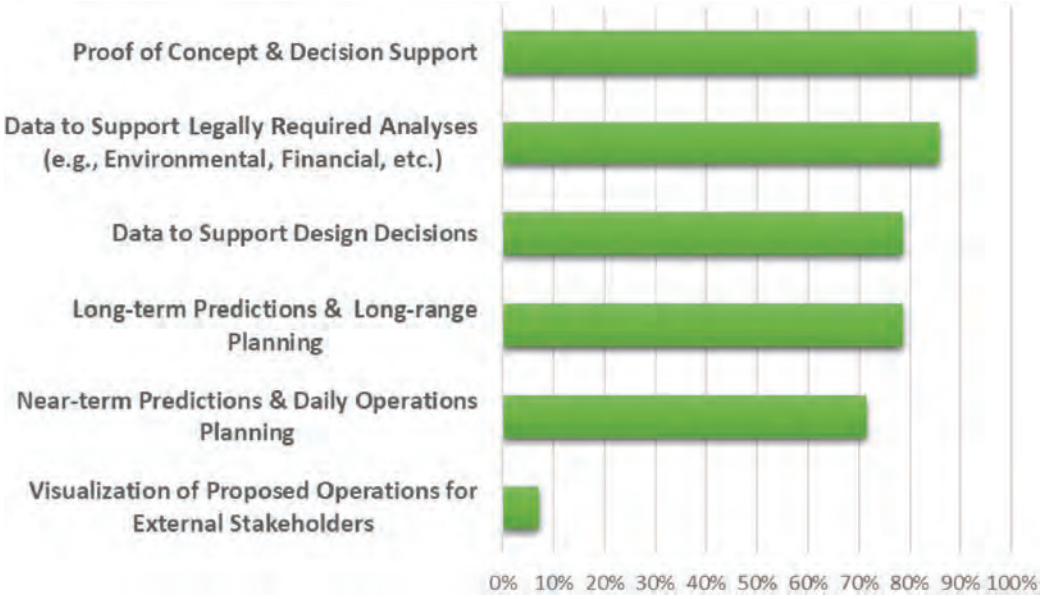


Figure 4. Survey results—simulation analysis benefits.

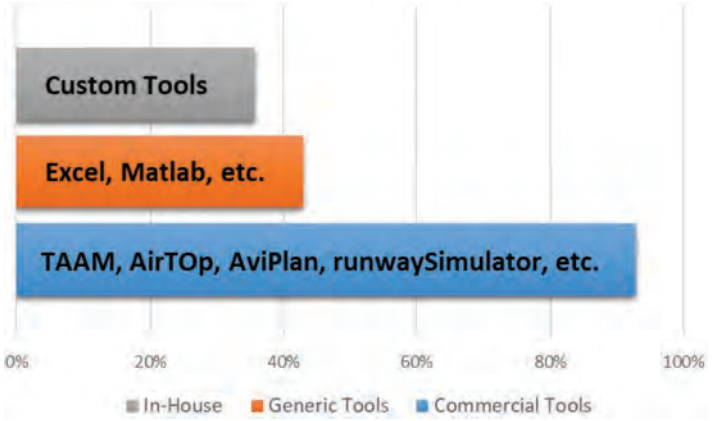


Figure 5. Simulation tool types.

Copyright National Academy of Sciences. All rights reserved.

Table 1. Software tools for airspace simulations.

Application	Software Tools
Master planning Procedure design Future concepts TMA/TRACON impact planning Obstruction planning Macro-economic analyses	ADSIM+ AirTOp Data Analysis Reporting Tool (DART) Future ATM Concepts Evaluation Tool (FACET) SIMMOD/SIMMOD Plus/Pro SkySafe Systemwide Analysis Capability (SWAC) systemwideModeler TAAM TARGETS

Other popular projects for the use of airspace simulation tools are air traffic control (ATC) procedure design and evaluation and Traffic Management Advisor/Terminal Radar Approach Control (TMA/TRACON) capacity planning, as well as obstruction analyses.

The most common airspace simulation projects and simulation tools that were noted by survey respondents and found in recent literature and project reports are listed in Table 1. Older and out-of-date simulation tools have generally been discounted for the purpose of this synthesis report unless they are narrowly focused and have few other alternatives.

As an example, Figure 6 depicts a TRACON simulation in AirTOp with vectoring aircraft on point-merge arrival procedures to two parallel runways.

Given some of the intricacies of ATC operations, the simulation tools required to model airspace operations are typically high-fidelity tools that require significant resources and hands-on knowledge. The inputs to airspace simulation models are often generated using more specialized tools that are used to develop Area Navigation/Required Navigation Performance (RNAV/RNP) procedures, airspace designs, and other air traffic flow models. For some of the common tools such as TAAM and AirTOp, there are direct input data conversion utilities available that can ease the interface burden. However, most of the time some manual data transformation and processing work is still required.

Table 2 provides a basic idea of the types of input data and external model interfaces that are commonly observed for airspace simulation studies.

Airport or Airfield

For detailed airport planning and design studies, airport or airfield simulation tools are most commonly used to analyze airport, runway, and gate capacities and associated delays. This report focuses solely on aircraft operations on the airfield. Numerous simulation tools can also simulate ground vehicles and other airfield objects, but these have not been specifically highlighted.

Presumably, since the fidelity of airfield simulation models and associated resource and data requirements can vary significantly from simple runway capacity spreadsheet models to more sophisticated custom discrete-event simulations of the entire airfield, the survey respondents generally indicated a slightly lower level of difficulty in conducting these types of simulation studies. However, higher fidelity simulation tools such as TAAM and AirTOp—both of which are capable of simulating airspace and an airport or airfield at the same time—still require significant knowledge and expertise.

Copyright National Academy of Sciences. All rights reserved.



Figure 6. AirTOp TRACON airspace simulation. Source: AirTopsoft.

Copyright National Academy of Sciences. All rights reserved.

Table 2. Airspace simulation inputs and external model interfaces.

Inputs	External Model Interfaces
Airspace procedure designs (i.e., RNAV/RNP)	Custom visualization tools
Airspace sectorization	Environmental models
Airspace route structures	Cost–benefit models
Airspace flow constraints	ATC tower simulators
Letters of Agreement (LOAs)	
Airspace/airport demand	

The types of studies most commonly associated with airport or airfield simulation tools are:

- Master plans (runway, gate, and airfield capacity and delay planning).
- Construction phasing and impact analyses.
- Irregular operations planning.

Airfield simulation tool fidelity can vary significantly depending on the phase of the study as well as the types of metrics that are needed to substantiate findings. For example, spreadsheet and high-level tools may be used during the initial or strategic planning phase, whereas high-fidelity tools are needed for master planning and comprehensive design assessments. Figure 7 shows a high-fidelity airfield simulation model of New York's John F. Kennedy (JFK) International Airport that may be used for detailed master planning studies.

There is quite a range of simulation tools that are currently used for airport and airfield simulation purposes, and the type of tool used depends heavily on the scope of the study. Table 3 lists the most commonly used airport or airfield simulation tools and their application areas.

Desired metrics also vary depending on the level of simulation tool fidelity required: from basic capacity-level information for runways and gates, to maximum airport and runway capacities and delays, to hourly throughputs, delays, gate usage, and other ground movement statistics (TransSolutions et al., 2014). Airport and airfield simulation studies may also support environmental assessment and impact studies.

Table 4 illustrates the types of input data as well as the simulation tool external interface outputs that are most common and relevant for airport planners.

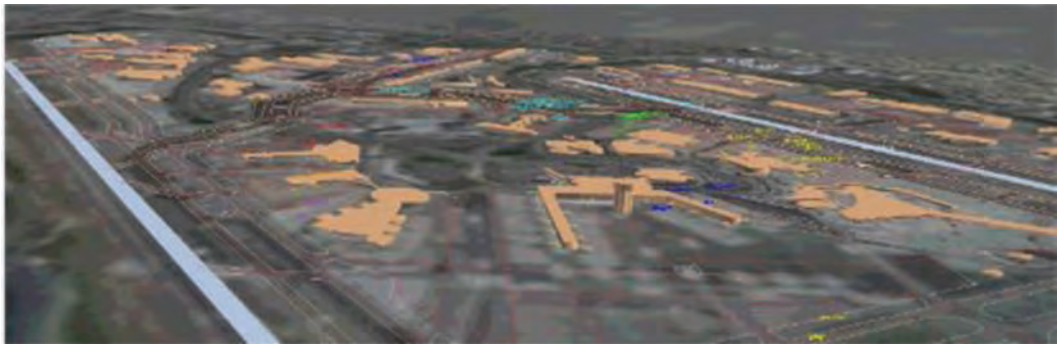


Figure 7. TAAM airfield simulation of JFK. Source: Cignus Consulting.

Copyright National Academy of Sciences. All rights reserved.

Table 3. Software tools for airport and airfield simulation and applications.

Application	Software Tools
Master planning	ACRP Spreadsheet Capacity Model
Capacity planning and design	AirTOp
Construction and operations planning	ArcPort
Resource planning	AviPlan Airside pro
Irregular operations planning	CAST Aircraft
Flight schedule evaluations	DART
	FAA/National Aeronautics and Space Administration (NASA) Annual Delay Model
	Future ATM [air traffic management] Concepts Evaluation Tool (FACET)
	MassMotion
	runwaySimulator
	SIMMOD SIMMOD Plus/Pro
	Systemwide Analysis Capability (SWAC)
	TAAM

Terminal Passenger Flow

Terminal passenger flow simulation tools can be used to model almost all passenger processes within and near the terminal building, including:

- Passenger enplanement and deplanement.
- Ticketing areas (desks, kiosks, remote).
- Security screening checkpoints.
- Baggage claim and carousels.
- Airport ground access and flows.
- Curbside processes.

These simulation tools follow a similar categorization as other tools with macroscopic, mesoscopic, and microscopic fidelity levels. A detailed discussion of the various types of passenger flow simulation concepts can be found in *Modeling Emergency Evacuation of Individuals with Disabilities* (Manley et al., 2011). Regarding simulation study complexity, survey respondents generally considered passenger simulations to be less difficult to conduct than airspace and airport or airfield studies.

The simulation analysts should be warned, however, that higher level, macroscopic analytical models of passenger flow within an airport terminal environment can be difficult to develop and

Table 4. Airport and airfield simulation inputs and external model interfaces.

Simulation Inputs	External Model Interfaces
Airfield design layouts	Custom visualization tools
Terminal design layouts	Environmental models
Airport future flight schedules	Cost–benefit models
Construction phases and zones	Gate schedule analysis tools
Airport ATC and ground operating procedures	

Copyright National Academy of Sciences. All rights reserved.

Table 5. Software tools for terminal passenger simulation.

Application	Software Tools
Terminal passenger capacity planning Passenger/terminal process design Check-in and security checkpoint design Staff/resource planning Emergency management and risk management Security threat analyses	ACRP models AirTOp ArcPort CAST Terminal MassMotion

analyze for terminal infrastructure capacity concerns due to highly complex terminal structures and transient demand patterns (Solak et al., 2009). Any study in which detailed statistics such as "*X percent of passengers have to wait more than X minutes*" are desired generally requires microscopic simulation tools. However, frequently the results obtained from these simulation tools are more detailed and substantive than the underlying assumptions actually support (Landrum & Brown et al., 2010).

Table 5 provides a list of passenger simulation tools and their application areas based on information gathered from the survey and recent literature.

To demonstrate visualization capabilities, Figure 8 presents a screenshot of a CAST Terminal passenger simulation model that integrates check-in and security checkpoint processes.

As *ACRP Report 25: Airport Passenger Terminal Planning and Design: Volume 1, Guidebook* stated:

> Computer simulations are often a useful tool in checking that each terminal passenger and baggage processing function is in appropriate balance with one another, thereby ensuring that no single function becomes a bottleneck that adversely affects the overall LOS (Landrum & Brown et al., 2010).

To support these types of analyses, metrics for passenger flow simulations can include arrival and departure patterns and trends, system capacities, resource availability, and security staffing requirements (Fayez et al., 2008). They may also focus on irregular operations that assess such



Figure 8. Comprehensive airport simulation technology (CAST) terminal passenger simulation. Source: Airport Research Center.

Copyright National Academy of Sciences. All rights reserved.

Table 6. Terminal passenger simulation inputs and external model interfaces.

Simulation Inputs	External Model Interfaces
Terminal design layouts	Custom visualization tools
Resource requirements (i.e., number of security checkpoints)	Environmental models
Passenger demand	Threat analyses
Boarding area designs and processes	Resource and staffing models
Check-in locations and strategies (i.e., kiosks)	
Emergency operational processes	

things as operational impacts of equipment failures, emergency evacuation and risk management strategies, and passenger flow system variability and randomness.

Table 6 provides an overview of relevant passenger flow simulation tool inputs and outputs.

Higher fidelity passenger flow simulation tools will accurately model passenger movements within a terminal structure using stochastic representations of historic passenger trends and relevant process times. Some tools may even use grids rather than directed passenger movement routes—or some dispersion of passenger movement along the route—to more accurately represent actual human behavior.

Terminal Systems

The modeling of terminal systems such as baggage systems and bus and train links generally follows a more generic queueing theory approach. Although some simulation tools may provide integrated passenger-process simulation capabilities, most studies typically depend on more generic medium-fidelity simulation software.

These types of tools simulate objects—baggage, buses, trains, and others—as they traverse through networks with capacitated resources. Most common simulation tools used to model generic processes—such as Arena, Vissim, and Simio—have both simple process-level visualization capabilities as well as 3-D representations of the same process. Table 7 lists the predominant software tools and applications.

As a visual example, Figure 9 shows a rail-link and bus terminal simulation using Simio.

Using metrics such as system and process delays and resource usage, these simulation tools can be used to assess process designs or, more tactically, to develop staffing and resource plans for day-to-day operations.

Table 7. Software tools for terminal system simulation.

Application	Software Tools
Baggage system planning Bus/train/shuttle planning	Arena Simio Vissim

Copyright National Academy of Sciences. All rights reserved.

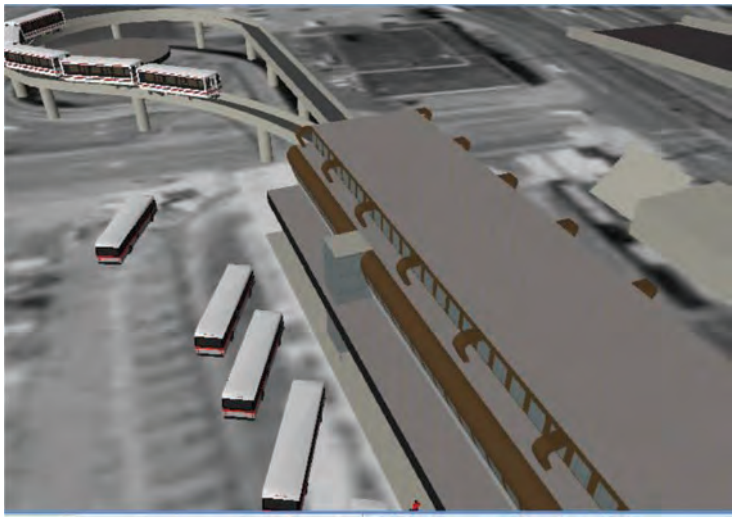


Figure 9. Simio rail-link and bus terminal simulation.

Ground-Access Simulation

Curbside operations are an integral part of the terminal operation as they are the interface between two transportation systems—airports and roadways. Like airside operations, ground access and curbside studies can focus on the evaluation of capital improvements such as roadway widening and construction of multiple curbside levels or on operational changes that aim to maximize vehicle throughputs or minimize delays. *ACRP Report 40: Airport Curbside and Terminal Area Roadway Operations* also provided a comprehensive summary of ground access and terminal area roadway analysis (LeighFisher et al., 2010). It was not until *ACRP Report 40* that guidelines that focused on the analysis of airport curbsides were defined. This report introduced a simulation tool called Quick Analysis Tool for Airport Roadways (QATAR), which was a simple macroscopic queuing model for curbside congestion and airport roadway analysis (Harris, Nourinejad, and Roorda, 2017).

The types of projects that curbside vehicle simulation tools generally support include:

- Airport road access planning.
- Airport terminal access planning.
- Work zone impact analyses.
- Consolidated transportation facility analyses.

Table 8 lists the predominant simulation tools and study areas based on survey results and searches of recent literature.

More microscopic curbside simulation tools are also available that focus on the simulation of individual vehicles on airport roadways. Although they can simulate curbside operations at a high level of detail, they are generally built on other urban transportation system simulation tools and require more detailed input data (Duncan and Johnson, 2000). These tools may not capture some of the intricacies of airport roadway systems, but their detailed vehicle simulation and visualization capabilities may be useful in graphically modeling curbside operations, which may still be useful in demonstrating and analyzing bottlenecks and vehicle ingress and egress issues (LeighFisher et al., 2010).

Copyright National Academy of Sciences. All rights reserved.

Table 8. Software tools for curbside and ground access simulation.

Application	Software Tools
Airport road access planning	ACRP tools
Airport terminal access planning	QATAR
Evaluation of construction zone impacts	Synchro
Toll plaza analysis	University of Missouri-St. Louis Queue Theory Model
Pedestrian flow analysis	Vissim
Transit facility planning	Vistro

Simulation Results and Application

The output data generated by simulation tools can vary significantly, as can the post-processing and result visualization capabilities. Many tools include custom analysis and reporting functionality that can present commonly used summary statistics, tables, and graphs. However, most experienced simulation users favor the exportation of raw output data to custom analysis tools that were purposely developed to obtain relevant results quickly and efficiently for incorporation into reports and presentations.

Simulation outputs can also, in many cases, be used to feed and support ancillary—or secondary—tools for further study. In addition to custom analysis packages, the types of secondary models that survey respondents noted most frequently include:

- Custom visualization and presentation tools.
- Environmental, noise, and emissions software.
- CBA or financial analysis tools.
- Others such as ramp and gate analysis charts and vehicle circulation models.

However, it should be noted that very rarely are simulation output data directly usable in secondary tools. Most of the time, data transformations and tweaks are necessary to ensure meaningful secondary tool results.

Multiruns

The survey also indicated that a clear majority of simulation analysts favor using multiruns for their airport and airspace simulation studies. Many passenger and generic process tools already natively use this Monte Carlo Simulation concept, and it is available as a function within most major airspace and airfield simulation tool packages. Multiruns allow airport and airspace modelers the ability to insert stochastic variability into the otherwise deterministic simulation software. Simulation scenarios are run numerous times with added randomness in factors such as:

- Aircraft, passenger, and vehicle departure and arrival times.
- Vehicle performance.
- Process times.

Essentially, the use of multiruns gives analysts the ability to conduct basic *sensitivity analyses* to assess how the modeled system behaves when environmental variables change. This adds more realism to the simulated scenarios and also provides validation of model robustness by ensuring that models can be run within a realistic range of environmental variables without gridlocking.

Copyright National Academy of Sciences. All rights reserved.

Simulation Model Maintenance

One aspect of applying simulation tools across the board that should not be discounted is that after a study has concluded, simulation models usually need to be maintained for future reuse. As such, they may require regular maintenance and input data updates in terms of:

- Airport, airspace, and passenger demand schedules.
- Gate and apron usage and selection strategies.
- Airport and terminal layouts.
- Terminal system processes, including locations of concessions and other subsystems that affect passenger flows.
- ATC procedures, taxi patterns, and other operational procedures.

Airport planners and designers considering the use of simulation tools should be aware that the maintenance of simulation models and the continual update and revalidation can require significant effort. This is particularly true with high-fidelity models.

Copyright National Academy of Sciences. All rights reserved.

CHAPTER 4

Simulation Tool Application Guidelines

It can sometimes be difficult to select the appropriate analytical method or simulation tool for a specific airport planning or design study. Some tools are custom designed for specific purposes, whereas others are more generic but equally as valid depending on the study. So, it is not only important to understand simulation tool capabilities but also where and how they may be used, what data and resources are generally required, and how difficult they are to apply. Using the information gained from the survey and from current industry practices, this report develops basic guidelines that airport planners and designers can use to select simulation tools.

Based on the survey, the factors that organizations generally considered in selecting simulation tools for their planning and design efforts focus on project type (i.e., master plans or capacity and demand studies), study budget, schedule, and required resources. More than 70 percent of respondents considered these factors to be important. Simulation input data requirements and availability and output capabilities also play a significant role but are considered less important. The industry and government acceptance levels as well as the level of anticipated study scrutiny and review seem to play only minor roles in the selection of simulation and analysis tools for studies. See Figure 10 for a comparison of the relative importance of each factor in simulation tool selection.

Based on this guidance, the project type was used as the primary decision factor for selecting a simulation tool for an airport planning and design study. The simulation tool selection is then further refined using important project characteristics. The survey and current industry trends indicated a basic selection process that can be described using the following ordered criteria:

1. Project Type.
2. Desired Metrics and Simulation Inputs and Outputs.
3. Resource and Schedule Requirements.

The fidelity of simulation tools used for airport planning varies significantly based on these criteria, as does the cost of the eventual construction project that is being considered. So, it is not surprising that detailed analytical and simulation models of airside operations are highly mature, given the considerable risk and cost of incorrect assessments. But, as Barnhart, Belobaba, and Odoni expressed, caution must be taken in the correct application of these models based on areas of analysis suited to the project (Barnhart et al., 2003).

Looking at the different kinds of projects that typically include simulation studies, the clear majority appear to focus on capacity and demand planning. Master planning and day-by-day operational planning projects also see frequent applications of simulation tools. Figure 11 provides a breakdown of simulation tool usage by project type based on survey data.

However, the simulation project types also vary depending on the area of focus: airspace, airport, terminal, or ground access. Table 9 lists the most common types of projects in which simulation tools were applied by area based on survey respondents and recent literature.

Copyright National Academy of Sciences. All rights reserved.

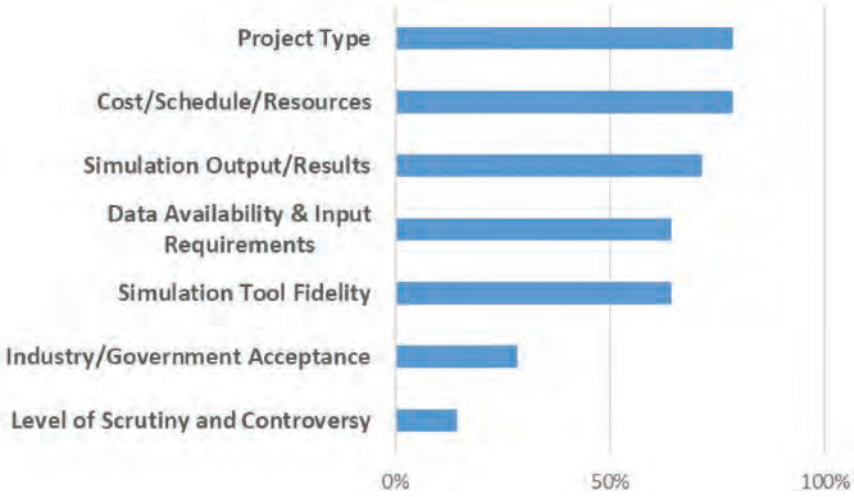


Figure 10. Survey results—factors considered in simulation tool selection.

The guidelines presented in this report will follow these basic criteria to assist airport planners and designers in selecting and applying simulation tools. The remainder of this section is organized by project type so that the reader can quickly navigate to areas of particular interest. Several matching case examples are presented for the reader to delve more deeply into simulation projects, requirements, and analysis metrics.

Master Planning

An airport master plan provides a blueprint for future operations from an operational and infrastructure point of view, including details of runway design, taxiway design, and terminal design. If the master plan is not correctly and carefully developed, airport performance can be affected for many years. The correct and appropriate use of simulation tools can significantly

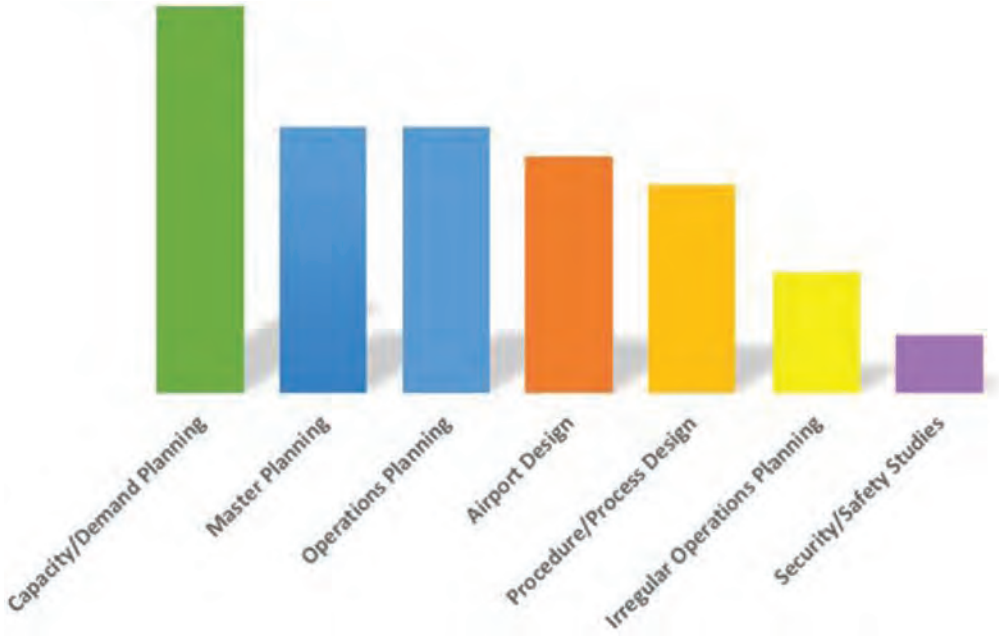


Figure 11. Survey results—simulation project types and usage.

Copyright National Academy of Sciences. All rights reserved.

Table 9. Simulation project types by area.

Airspace	Master planning Procedure design Future concepts	TMA/TRACON impact planning Obstruction planning Macroeconomic analyses
Airport/Airfield	Master planning Capacity planning and design Construction and operations planning	Resource planning Irregular operations planning Flight schedule evaluations
Terminal Passenger	Terminal passenger capacity planning Passenger/terminal process design Check-in and security checkpoint design	Staff/resource planning Security threat analyses
Terminal Systems	Baggage system planning Bus/train/shuttle planning	
Ground Access and Curbside	Airport road access planning Airport terminal access planning Evaluation of construction zone impacts	Toll plaza analysis Pedestrian flow analysis Transit facility planning

assist in informing key decision making in this regard. However, a high-fidelity and expensive simulation software used on a small project can easily drive project costs and schedules out of control. Likewise, a low-fidelity but cheaper tool may not be sufficiently capable of accounting for complex scenarios that require consideration on a large project. Therefore, picking the right simulation tool for the project at hand is the key factor in achieving a successful master plan.

For example, for an airfield capacity study in which delay numbers are not important but simple runway throughput is needed (i.e., a general aviation airport with one runway or a proposed new runway), runwaySimulator may be an appropriate simulation tool. It can be considered a low- to medium-fidelity simulation tool, but at the same time the tool is sufficiently accurate to assess runway capacity without any ground constraints. The input requirements for runwaySimulator and other low-fidelity simulation tools are minimal and associated preparation time is hours versus days as compared with higher fidelity models such as TAAM, SIMMOD, or AirTOp (LeighFisher et al., 2012).

As expected, medium-fidelity tools such as the Airport Delay Simulation Model (ADSIM) require more extensive input data to achieve greater accuracy in modeling aircraft and analysis data. Although this increases the development time for simulation scenarios, they deliver more robust and detailed results. Table 10 illustrates the use of ADSIM and the Runway Delay Simulation Model (RDSIM) within an initial master planning capacity study at Memphis International Airport in Tennessee.

Figure 12 shows the ADSIM simulation user interface of an airfield, including runways, taxiways, aprons, and gates.

For more complex master plans, particularly for airports within metropolitan areas, an airspace component may force the use of high-fidelity simulation tools. If a study calls for new runways, major airfield reconfigurations, or significant airspace changes, higher fidelity models such as AirTOp, TAAM, SIMMOD, or CAST Aircraft are more appropriate simulation tools. Specifically, if an environmental impact statement (EIS) or environmental impact assessment (EIA) is planned or anticipated, a high-fidelity model is the best choice. Although these simulation tools require the most effort and resources in terms of data processing, preparation, setup,

Copyright National Academy of Sciences. All rights reserved.

Table 10. Medium-fidelity airport master plan simulation case example.

Study Purpose	Capacity Enhancement Plan Update–1997
Organization	Federal Aviation Administration Technical Center and Memphis-Shelby County Airport Authority
Environment	An example of dozens of such Capacity Enhancement Plans prepared by the FAA Technical Center and local Airport Capacity Design Teams. Airfield capacities and aircraft delays were calculated for baseline conditions (existing demand and operating procedures). Benefits of proposed improvements were measured against these baseline capacities and delays.
Model Used	ADSIM and RDSIM
Metrics	Hourly airport capacity corresponding to an average aircraft delay of 4 minutes total and average aircraft delays and costs.
Resource and Data Requirements	Airfield layouts, operational procedures, and rules Weather conditions Hourly, daily, and annual traffic data and mix
Applicability to Airport Planning/Design	Airfield (runway) capacity planning
Reference	*ACRP 03-17: Master Plan Update, Memphis International Airport*, LeighFisher, 2008, available at: http://onlinepubs.trb.org/onlinepubs/acrp/docs/ACRP03-17_FR.pdf. P. 1-21.

and calibration, they provide the necessary granularity and environmental controls to support costly decisions.

As mentioned in the Chicago O'Hare International Airport (ORD) Final EIS document, the rationale for using high-fidelity models for complex master planning includes their ability to facilitate direct and real-time review of simulation models by air traffic controllers and other subject matter experts (FAA, 2005). This same rationale was also a driving factor in the use of TAAM for the Philadelphia International Airport (PHL) Capacity Enhancement Project, as outlined in Table 11. The PHL TAAM simulation study also fed information to the subsequent EIA and EIS.

Large hub airports across the globe have used high-fidelity simulation tools to drive investment decisions for many years. As expected though, the simulation know-how, input data requirements, and time required to prepare simulation scenarios can be quite extensive.

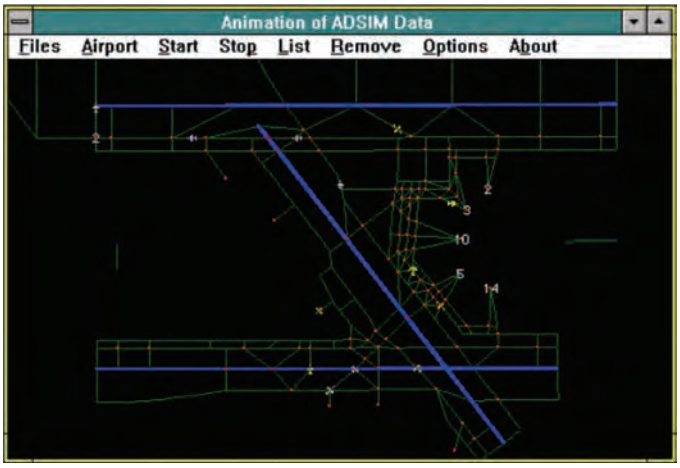


Figure 12. ADSIM airport simulation. Source: FAA.

Copyright National Academy of Sciences. All rights reserved.

Table 11. High-fidelity airport master plan simulation case example.

Study Purpose	Philadelphia International Airport Capacity Enhancement Project
Organization	Philadelphia International Airport (PHL)
Environment	To enhance airport capacity and accommodate current and future aviation demand in the Philadelphia metropolitan area during all weather conditions
Model Used	TAAM
Metrics	Capacity, delays
Resource and Data Requirements	Detailed future airport layouts, including construction years DDFS (current and future), Standard Instrument Departure (SID) and Standard Terminal Arrival Routes (STARs) Taxiway flow diagram, etc. Current and future airport gate and apron usage
Applicability to Airport Planning/Design	Airport master planning
Reference	*FAA Record of Decision for Capacity Enhancement Program at Philadelphia International Airport*, 2010.

Some of the most common metrics derived from such tools are:

- Capacity.
- Delays.
- Conflict points.
- Facility utilization.

Delays are often annualized as part of a CBA in which return on investment decisions can be made that measure monetary benefits of improved future capacity and efficiency against the cost of capital improvement projects.

Construction Phasing Plans and Impact Studies

The simulation of construction phasing plans goes hand in hand with the master planning process. As preferred alternative layouts or operational scenarios are identified, construction time and resource availability dictate when parts of the master plan layout are going to be constructed. This generally results in a construction phasing plan in which each phase has specific impacts on airfield or terminal operations that require impact assessments.

For this task, planners are likely to use the same simulation tools that were used throughout the initial master planning effort because they have been validated and represent current and future demand-driven operational scenarios that only need to be adapted. Existing master plan models are generally used and modifications are made to represent the various construction phase layouts, operational restrictions, and proposed taxi paths and gate assignments. Frequently, the simulation tools can also be used to optimize taxi flows and ground movements.

In contrast to the master plan simulation effort, the metrics will focus more on ground operations efficiency and delays than airport-level capacities and future capacity and demand issues. Although it is important to ensure that ground traffic restrictions do not starve the runways, the analysis will likely focus more on airfield delays, choke points, and demand variability.

Copyright National Academy of Sciences. All rights reserved.

Airport Capacity and Delay Studies

Whereas master planning studies use purpose-built simulation scenarios to evaluate proposed airport layout changes, capacity and delay studies are commonly used to assess the impacts of future demand growth on existing or proposed infrastructure. These studies are more common and informal than the ones for master plans and typically evaluate aspects related to future airport capacity saturation as well as benefits gained from major infrastructure changes such as new runways and terminals.

It is common for airport capacity and delay studies to use a series of synthetic future demand schedules to determine the capacity of a given airfield or terminal layout. They can focus on any aspect of the airfield, including runways, taxiways, gates, or the entire airfield itself. The scope of the study naturally dictates the fidelity level of the simulation tool that should be used.

Runway capacity studies simply focus on arrival and departure movements on runways and can take runway dependencies, wake turbulence separation criteria, and basic runway allocation rules into consideration. Using future demand schedules, various alternative runway capacities can be determined for airfield operating configurations, including additional runways, the closure of runways, or the reallocation of traffic by runway.

These types of studies may best be conducted using low- to medium-fidelity simulation tools such as runwaySimulator or Airtopsoft's runway capacity module. The use of runwaySimulator for basic airport or runway-level capacity planning is outlined in Table 12, and a basic visualization of runway operations is illustrated in Figure 13.

Often, runway capacity studies are conducted as initial phases of larger airfield or airspace planning studies and they discount other airfield capacity bottlenecks. As the case study in Table 12 illustrates, runwaySimulator is a popular tool for making simple airport capacity predictions across multiple operating configurations, weather conditions, and environments.

When taxiway, apron, or gate capacity is a perceived problem, the scope of the simulation effort naturally increases to a point where a more detailed simulation of the entire airfield may be required. At this stage, higher fidelity models such as AirTOp, TAAM, SIMMOD, CAST Aircraft, and RAMS come into play for more detailed modeling of the airfield. Table 13 outlines

Table 12. Runway-level capacity planning simulation case example.

Study Purpose	FAA's Airport Capacity Benchmark Report
Organization	MITRE
Environment	Simulation studies in support of the development of airport capacity prediction benchmarks that used the predominant operating configurations of some of the busiest airports in the United States. This includes current airport layouts as well as any known future capital improvements.
Model Used	runwaySimulator
Metrics	Airport and runway capacity by configuration
Resource and Data Requirements	Current and future airport layouts Weather conditions Airport configuration and historic usage Airport facility reported arrival and departure rates (for validation) Fleet mix
Applicability to Airport Planning/Design	Airport runway system capacity planning
Reference	Kuzminski, 2013

Copyright National Academy of Sciences. All rights reserved.

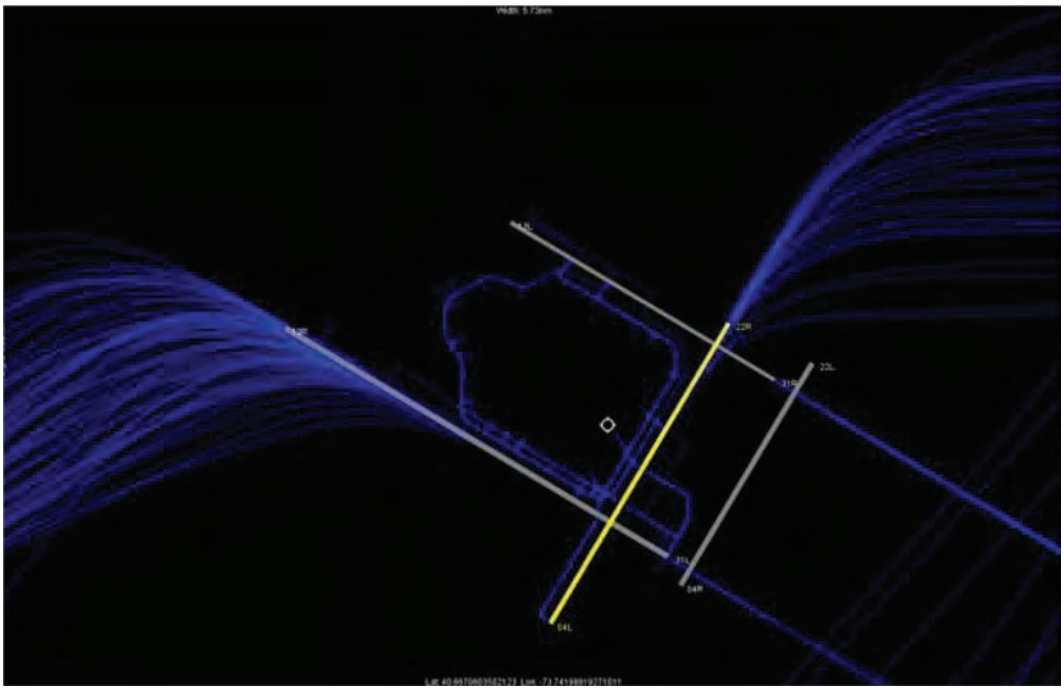


Figure 13. Simulation of John F. Kennedy International Airport using runwaySimulator. Source: FAA.

a higher fidelity simulation study of airport and runway capacity in which other factors such as taxiways, aprons, and gates are known to have had direct impacts on the overall airport and runway capacity.

For higher fidelity airfield capacity and delay studies, the simulation effort mimics that of a master plan, but generally, capacity and delay studies focus more on throughput and delay metrics based on specific future demand rather than on the comparison of different layout alternatives.

Table 13. Runway and airport capacity simulation case example.

Study Purpose	New Airfield (Runway) and Capacity (London–Heathrow Airport), 2014
Organization	Gatwick Airport Ltd./DFS
Environment	Simulations that assessed the operational benefits of a third runway at London–Heathrow International Airport (LHR). LHR is one of the world's busiest and most efficient airports. This simulation study supported a larger capacity-demand study.
Model Used	TAAM/AirTOp
Metrics	Runway capacity
Resource and Data Requirements	Airport and runway layouts Flight schedules Arrival and departure procedures Runway dependencies and wake separations
Applicability to Airport Planning/Design	Runway capacity planning
Reference	DFS Deutsche Flugsicherung GmbH, Jan. 2015

Copyright National Academy of Sciences. All rights reserved.

Terminal Passenger Flows

A passenger's first impression on their journey is the airport terminal. From the entrance to the boarding gate, the airport terminal is designed to ensure a seamless passenger experience from check-in through security checkpoints to the boarding areas. Airport terminals also offer commercial and entertainment facilities that passengers encounter and interact with along their travel paths. Among others, terminal design objectives generally focus on maximizing passenger throughput and minimizing queues at check-in and security screening points. Issues with terminal passenger flows can cause increased security threats, delays of connecting flights, and passengers missing their flights.

An ever-present desire to provide efficient and enjoyable passenger flows and experiences within the terminal, combined with the need to maximize commercial and other revenue-generating facilities, creates a popular opportunity to apply simulation tools for both terminal design and operational planning purposes. The survey indicates that most airport planners use passenger flow simulation within the terminal structure primarily for the following types of studies:

- Terminal passenger capacity planning.
- Security checkpoint design and resource planning.
- Check-in process capacity planning.

About 50 percent of survey respondents indicated that they use simulation tools to directly influence terminal design decisions. However, it should be noted that there is no mandated use of passenger flow simulation tools for airport terminal planning.

Simulation tools that focus on terminal passenger flow planning can provide significant value to the overall airport planning and design process. Such tools can assist in the definition and assessment of passenger paths both within and outside the terminal structure. They can also be used to discover passenger flow bottlenecks and to design adequate emergency evacuation routes. Passenger simulation tools are popular in industry because of their applicability to any number of other studies that evaluate passenger traffic. Relatively generic tools such as MassMotion, Arena, and Simio can not only be used to model airport terminal passenger traffic but also to model other areas such as train terminals and shopping malls.

As the example in Figure 14 shows, the MassMotion terminal passenger simulation tool provides detailed visualizations of passenger movements that are supported by analytical capabilities that allow planners to analyze and fine-tune design decisions.

Most generic passenger flow simulation tools can be tailored to specific airport environments, but the processes, queues, and constraints need to be well defined and understood. Table 14 illustrates the use of Arena to simulate check-in and security checkpoint queues using a combined flow model.

Several planners also indicated that they use simulation tools for emergency evacuation and response scenarios. Table 15 presents a case example in which an agent-based model called BUMMPEE was used to simulate airport emergency evacuation procedures for disabled individuals.

Some simulation tools such as AirTOp—which is primarily an airfield and airspace simulation tool—also provide modules that simulate terminal passenger flows. The benefit of using these higher fidelity tools for passenger flow analyses is that they may be readily linked to airfield models to provide a more holistic, integrated assessment of the entire interconnected airport system.

Copyright National Academy of Sciences. All rights reserved.

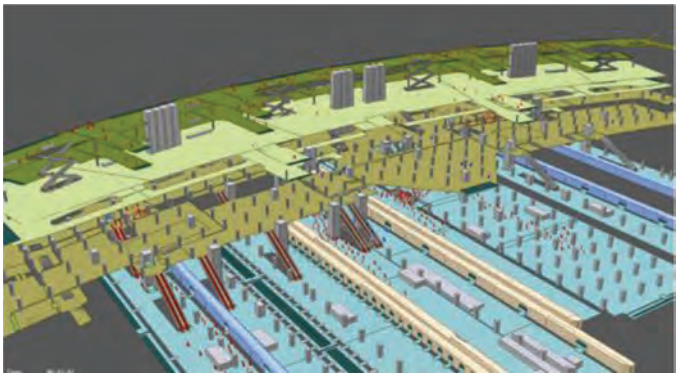


Figure 14. MassMotion terminal passenger simulation. Source: Oasys Software at http://www.oasys-software.com/products/engineering/massmotion.html.

Generally, passenger flow simulation tools use stochastic distributions of passenger transit and processing times to represent real-world processes. These distributions of historic passenger characteristics, behavior, and trends are based on data collected by the airport or through visual observation conducted purposely for the studies. Combined with information on capacitated queues and resources—such as check-in counters and security checkpoints—they represent the basic data requirements for terminal passenger simulations.

As one might expect, the analysis metrics are geared toward processing times, delays, and resource usage. Some examples of passenger terminal flow simulation metrics are:

- Passenger time in terminal.
- Check-in and security wait delays.
- Use of security checkpoints and staff.

The results from these simulations can be used to assess terminal design decisions (i.e., passenger flow bottlenecks) as well as to optimize staffing and resource plans for objects such as security personnel and airline check-in counters.

Table 14. Check-in process and security checkpoint case example.

Study Purpose	A Discrete Event Simulation to Model Passenger Flow in the Airport Terminal
Organization	Naples International Airport (Naples, Italy)
Environment	With a focus on check-in and security checkpoint processes, this study examined the use of simulation tools to improve the efficiency of check-in and security checkpoint procedures inside the airport terminal.
Model Used	Arena
Metrics	Hourly queue waiting time Number of users in the queue
Resource and Data Requirements	1. Passenger arrival rate 2. Service time 3. Queue length
Applicability to Airport Planning/Design	Check-in and security planning
Reference	Guizzi et al., 2009

Copyright National Academy of Sciences. All rights reserved.

Table 15. Emergency evacuation and risk management case example.

Study Purpose	Modeling Emergency Evacuation of Individuals with Disabilities in a Densely Populated Airport
Organization	Transportation Research Board case example
Environment	Passenger terminal emergency evacuation study focused on the impacts of procedures and different passenger segments on terminal design
Model Used	Agent-based model (BUMMPEE)
Metrics	1. Limitations of the pier airport design during emergency evacuations 2. Identification of the individuals most at risk or those with lower stamina 3. Potential bottleneck areas
Resource and Data Requirements	Terminal layouts Passenger demand information Population distributions of different passenger segments, including the nondisabled, motorized wheelchair users, visually impaired, etc.
Applicability to Airport Planning/Design	Evacuation planning
Reference	Manley et al., 2011.

Terminal System Design and Planning

In addition to passengers, terminal operations depend on the efficient operation of several other subsystems within the terminal structure. Passenger baggage systems are an integral terminal subsystem that connects passengers and their baggage with aircraft. Inefficiencies in this process can cause large-scale airport delays. There are several other processes that were mentioned frequently by airport planners as focus areas for simulation studies. Studies focused on interterminal train links and bus connections are also perfect candidates for the use of simulation tools. The majority of these types of studies use generic simulation tools that simply model processes using transit and process times, queues, and demand distributions with varying characteristics. Table 16 presents a case example of Riga International Airport's baggage handling system using a simulation tool called ExtendSim.

Table 16. Baggage process simulation case example.

Study Purpose	Riga International Airport Baggage Handling System Simulation
Organization	Transport and Telecommunication Institute
Environment	An increase in travel demand at Riga International Airport in Latvia requires an assessment of the current check-in and baggage system processes to ensure sufficient capacity for future demand growth.
Model Used	ExtendSim
Metrics	System utilization Queue length for check-ins
Resource and Data Requirements	Airport infrastructure (check-ins, conveyer length) Flight schedule (present and future) Aircraft and airline load factors Peak arrival times Check-in start and end time
Applicability to Airport Planning/Design	Check-in and baggage handling system planning and management
Reference	Savrasovs et al., 2009

Copyright National Academy of Sciences. All rights reserved.

Similar to passenger flow simulation tools, baggage system simulations depend heavily on focused input data. This includes daily passenger demand distributions, flight and gate schedules, the percentage of passengers that have checked versus carry-on baggage, the percentage of checked bags that undergo manual Transportation Security Administration checks, the percentage of transiting versus origin-destination passengers, and so on.

The analysis metrics from these terminal process simulation studies are customized depending on the analysis scope and methodology. For baggage system simulations, airport planners may generally be interested in average and maximum bag time in the system, flights delayed due to late baggage arrivals, etc. Simulations of train and bus links may, however, focus more on passenger wait times in support of resource optimization and schedule planning studies.

Ground-Access Studies

Airport terminal curbsides represent critical interfaces between standing vehicles, moving vehicles, and pedestrians. They act as the capacity buffer between the road delivery system and the airport terminal building and represent the primary terminal access interface for passenger drop-off and pickup, including emergency access. The correct design and dimensioning of curbside areas is crucial for the efficient movement of passengers in and out of the terminal area.

Although not as popular a focus for simulation studies, several airport planners noted that they frequently use various simulation tools for the planning, design, and operational impact analysis of curbside operations. Based on the survey responses, QATAR and Vissim are the most popular simulation tools for curbside operations. Some tools even have capabilities for roadway and intersection modeling, which can be used to assess the road interfaces between airport roadways and the highway infrastructure.

A recent master plan update for Los Angeles International Airport used Vissim to simulate curbside and road operations to assess the traffic impacts of a new terminal concourse. Table 17 provides details for this example.

Table 17. Ground-access vehicle simulation case example.

Study Purpose	Los Angeles International Airport master plan
Organization	Los Angeles World Airport
Environment	An assessment of potential traffic-related impacts associated with new facilities proposed as a part of the Midfield Satellite Concourse and the Central Terminal Processor
Model Used	Vissim (curbside, roadway)
Metrics	Level of service
Resource and Data Requirements	Traffic data for peak hour Traffic mix (heavy and nonheavy ratio, etc.) Future traffic conditions Aircraft/Airline load factors Airline passenger (PAX) schedules Vehicle counts and classifications Parking structure vehicle count
Applicability to Airport Planning/Design	On-airport surface transportation system
Reference	*Los Angeles International Airport–Midfield Satellite Concourse, Draft EIR*, available at: https://www.lawa.org/uploadedFiles/OurLAX/pdf/4_6_MSC_DEIR_On-Airport_Transportation.pdf.

Copyright National Academy of Sciences. All rights reserved.

In many ways, the simulation tools used to model curbside operations are based on generic simulation principles that focus on queues and processing times. Data input requirements for curbside simulation tools generally include information on the roadway layout, vehicle traffic demand, vehicle traffic characteristics, and passenger and group characteristics, as well as airline and flight information for passengers to ensure that pickup and drop-off areas are used appropriately.

As with other generic simulation tools, the analysis metrics will focus on the achieved level of service, delays, and throughput capacities.

Airspace Procedure Design

Airspace procedural changes are a necessary companion to airport studies that involve new runways. NextGen Performance Based Navigation (PBN) procedures focus on improving flight efficiency and reducing carbon footprints. Other factors such as airport and airspace capacity, safety, and environmental impacts also need to be considered when new airport arrival or departure procedures require evaluation.

Simulation tools are frequently used in the design and evaluation phases of airspace procedure design studies. Terminal Area Route Generation and Traffic Simulation (TARGETS) and Instrument Approach Procedures Automation (IAPA) are the most commonly used procedure design tools and generally require specialized knowledge and expertise due to their focus on air traffic control operations and procedures. Although the proposed procedures are tested for flyability within these specialized software tools, the tools cannot assess flight efficiency, airspace impacts, conflicts, or economic benefits. This is particularly true in metropolitan environments where multiple large- and medium-size airports operate in close vicinity and procedures need to be internally optimized for flight profiles, but also deconflicted with other airport procedures at the same time.

Several high-fidelity simulation tools are often used for this assessment, including AirTOp, TAAM, and SIMMOD. For example, FAA's New York Airspace Redesign effort used TAAM to test and evaluate new RNAV/RNP procedures, route structures, and airspace sectorizations. This study explored operational impacts and supported subsequent environmental assessments. FAA and MITRE also used AirTOp as the simulation tool of choice for an airspace and procedure design feasibility analysis of a new proposed airport in the Chicago metropolitan area. More detail on this example is provided in Table 18.

The benefit of using high-fidelity simulation tools for these assessments is that they can add a level of realism in terms of dynamic flight schedules—or demand—as well as simulate airport and airfield operations, runway constraints, traffic flow management initiatives, and other en-route airspace restrictions.

Some of the common metrics that are extracted from simulations for airspace procedure design studies include:

- Airport and runway capacity.
- Transition (SID/STAR) fix capacities.
- Conflict points between procedures.
- ATC sector throughputs or capacities.
- Flight distances on procedures.
- Climb and descent profiles on procedures.
- Fuel burn and costs.

Copyright National Academy of Sciences. All rights reserved.

Table 18. Airspace procedure design simulation case example.

Study Purpose	Initial Airspace and Procedures Feasibility Analysis for the Proposed South Suburban Airport (SSA)
Organization	FAA and MITRE
Environment	The project's primary objective was to examine the existing Chicago metropolitan area air traffic and airspace structure to determine the feasibility and challenges of integrating a supplemental commercial airport (SSA) into the existing structure.
Model Used	AirTOp
Metrics	Throughput by airport Ground delay by airport Time in level flight by airport Flight time by airport Potential conflicts requiring ATC resolution Prearranged coordination climb operations involving new procedures Distance in level flight for proposed SSA departures
Resource and Data Requirements	Current SIDs, STARs, airspace, and runway operations at ORD and MDW Proposed SSA RNAV SIDs and STARs using TARGETS software Proposed airspace design changes Varying demand and schedule scenarios East and west flow scenarios
Applicability to Airport Planning/Design	Airspace procedure design and planning ATC impacts
Reference	*Initial Airspace and Procedures Feasibility Analysis for the Proposed South Suburban Airport (SSA)*, available at: http://www.southsuburbanairport.com/MasterPlan/reports/MP-Report-Airspace.htm.

Simulation tools can also feed noise and emission models that use flight tracks, traffic demand, and traffic mix information to visually depict changes in ground noise and emission impacts. This information can support subsequent EISs as well as public outreach efforts.

Air Traffic Management Concept Evaluation

A common question that airport planners frequently face is whether an airport is NextGen-capable. Technologies such as PBN, arrival and departure management, 4-D trajectory-based operations, information sharing, and enhanced ATC surveillance are some of the NextGen concepts that airports are routinely preparing to accommodate. Requirements for operational improvements are continuously requested by airlines and business aircraft operators to reduce delays and drive profits. These improvements can be focused on the airport surface, such as with advanced departure managers, or in the airspace as with PBN procedures, continuous descent and climb operations, and the like.

For example, at San Francisco International Airport FAA evaluated and implemented closely spaced parallel operations using TAAM as their simulation tool of choice. Closely spaced parallel runways are spaced less than 2,500 feet apart and are authorized for 1.5-nautical mile dependent staggered approaches. FAA continued to use simulation tools to assess these benefits and eventually implemented closely spaced parallel operations in Boston, Cleveland, Memphis, Newark, Philadelphia, Salt Lake City, San Francisco, and Seattle.

NextGen technology-focused simulations require input data that are fairly detailed and frequently more labor intensive to process. The primary reason for this is simply that simulation

Copyright National Academy of Sciences. All rights reserved.

Table 19. Future concept case example.

Study Purpose	Assess Benefits of CDOs at Atlanta International Airport
Organization	Embry-Riddle Aeronautical University
Environment	A flight efficiency comparison of existing arrival procedures (STARs) at Hartsfield–Jackson Atlanta International Airport and proposed RNAV/RNP Continuous Descent Operations procedures with two-dimensional (2-D) track freedom.
Model Used	TAAM
Metrics	Flight time, fuel cost, ATC conflicts
Resource and Data Requirements	Traffic demand schedule, fleet mix, current and proposed SID and STAR configurations, route structures and use of arrival and departure fix
Applicability to Airport Planning/Design	Future ATC procedure concept evaluation
Reference	Wilson and Hafner, 2005.

studies in support of NextGen technology evaluations usually require the use of multiple simulation tools and data input/output interfaces must be supported. This is particularly true because simulation tools are rarely purpose-built to be able to directly model future concepts and technologies. For example, simulation studies that focus on the benefits of advanced arrival managers will likely require the use of other software tools to preprocess flight schedules and provide advanced arrival sequences that can then be integrated and simulated in other high-fidelity tools such as AirTOp, TAAM, or SIMMOD.

Table 19 demonstrates the use of TAAM for a conceptual study on the benefits of Continuous Descent Operations (CDOs) at Hartsfield–Jackson Atlanta International Airport.

Another popular tool for the analysis of systemwide impacts of future air traffic management concepts and technologies is NASA's FACET simulation tool. FACET was designed to focus more on the National Airspace System (NAS) and macroscopic impacts of future technologies than on individual airports or airfields and facilities. So, it may be better suited to simulating impacts of airspace-related future NextGen and Single European Sky ATM Research (SESAR) concepts such as Arrival Manager/Departure Manager (AMAN/DMAN) and 4-D trajectories. This is similarly true for DART, which is a newer simulation tool that can very quickly model aircraft across the entire NAS. DART uses weather data (convective and nonconvective) in addition to wind information for simulation and analysis purposes.

Because simulations of future concepts are typically conceptual and conducted at a high level, the analysis metrics are generally geared toward assessments of airspace and ATC impacts, ATC and runway throughput capacities, and flight efficiency.

Depending on the concepts and technologies being evaluated, it is not uncommon to simulate future operations using high-fidelity simulation tools with existing master planning or capacity and delay models to increase the realism of the overall assessment.

Metrics and Analysis

Most airport planners and designers—more than 85 percent based on the survey—use simulation tools for strategic, long-term planning projects. These types of projects are generally driven by capital improvements in which simulation results are used to justify design and construction decisions.

Copyright National Academy of Sciences. All rights reserved.

However, more than 50 percent of the survey respondents also noted that they frequently use simulation tools for more tactical operations planning tasks. The types of projects noted include:

- Near-term improvements that assume the use of existing procedures and technology.
- Baggage systems and work zone impacts.
- Construction planning and TFM initiatives.
- Construction phasing and specific capacity.
- Operational planning.
- On-call planning work that requires simulation tools to complete daily tasks.

These types of studies may require different analysis methods and usually have shorter turn-around times because they impact near-term operations.

Metrics obtained from these simulation tools can generally be classified as:

- Summary statistics.
- Dynamic demand-driven graphs and tables.
- Visualizations of simulated traffic.

As Figure 15 shows, the vast majority of airport planners that responded to the survey use detailed simulation metrics and derived Excel spreadsheet analysis tools in their analyses and

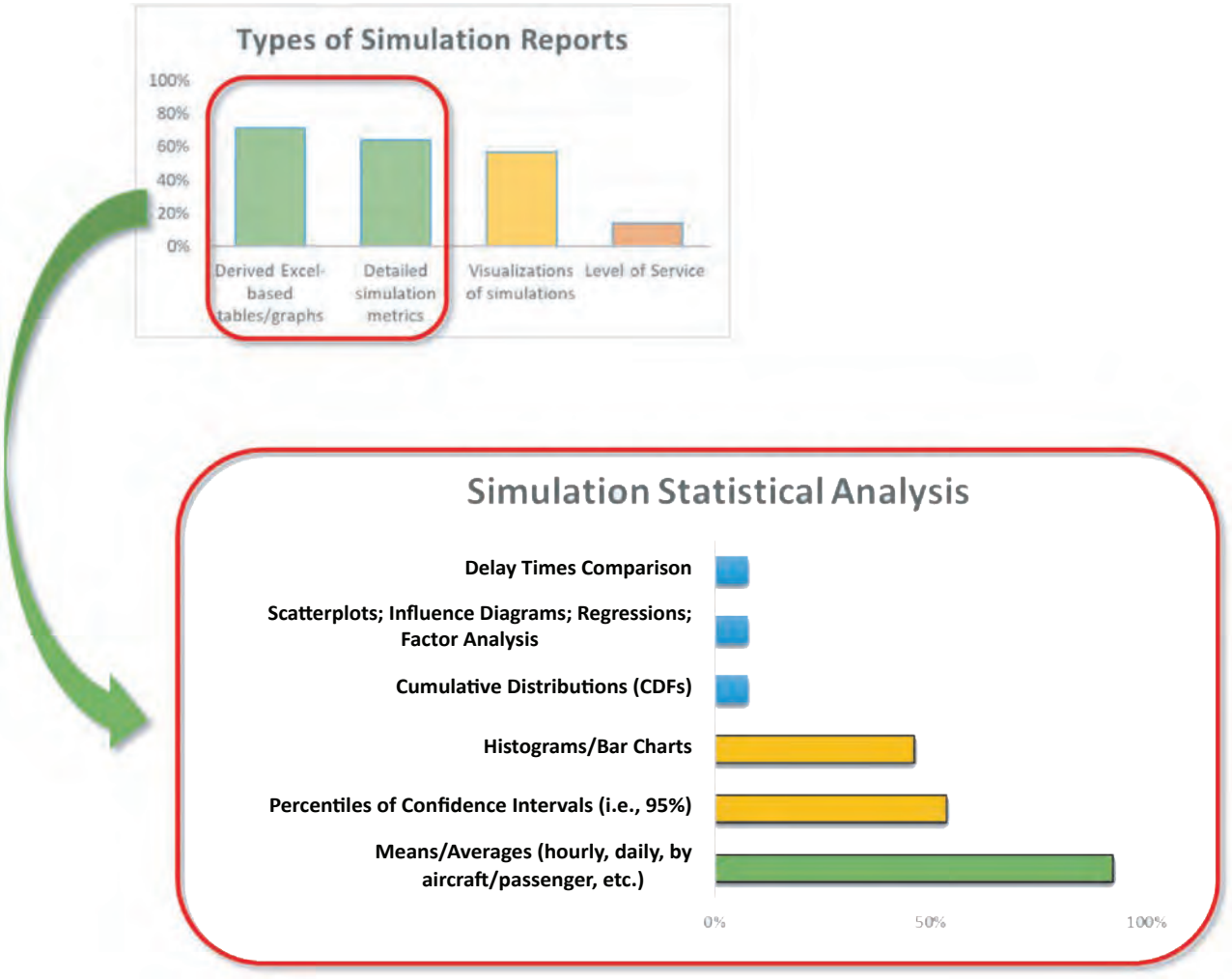


Figure 15. Survey results—simulation metrics and statistics.

Copyright National Academy of Sciences. All rights reserved.

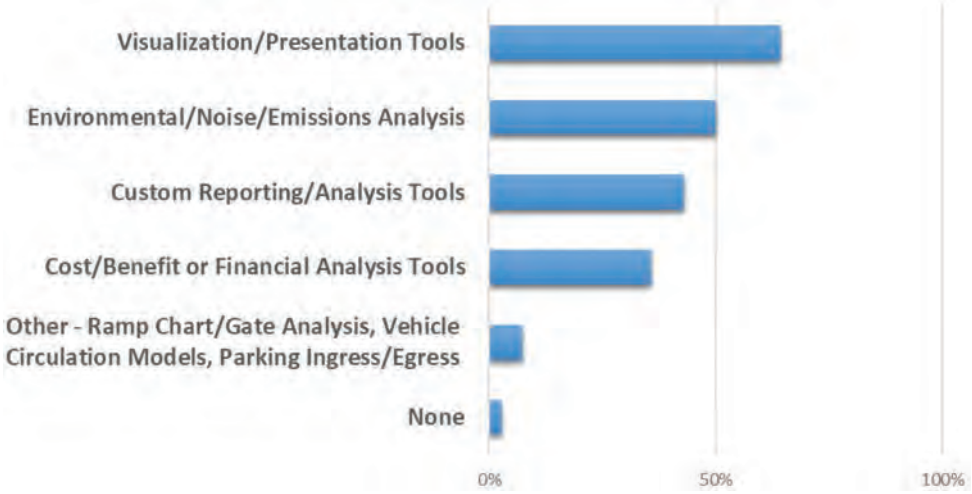


Figure 16. Secondary models using simulation output data.

report presentations. Frequently, the need for more detailed metrics may require custom software tools to extract and analyze simulation data. Metrics are then summarized using basic means and averages (e.g., daily, hourly, or by aircraft and passenger), confidence intervals, or standard histograms.

Simulation tool outputs are also frequently used to support other applications and tools. These ancillary models range from more sophisticated visualization and analysis tools—including statistical and financial—to models that support more detailed analyses of secondary factors such as noise, emissions, gating plans, or vehicle and parking ingress and egress. Figure 16 lists all of the secondary tools and models noted by survey respondents.

An important aspect of simulation analysis that should not be overlooked, however, is that of study and result validity and reliability. The majority of airport planners or organizations generally find simulation results to be reliable and truthful. However, several noted that the reliability of simulation study results is directly dependent on factors such as

- Input data availability and quality.
- The supporting simulation and study methodology.
- The analysts.
- Stakeholder participation and agreement.

Copyright National Academy of Sciences. All rights reserved.

CHAPTER 5

Simulation Tool Characteristics and Capabilities

This chapter provides readers with relevant information on each airspace, airfield, terminal, and ground-access simulation tool mentioned previously. The information presented for each tool is still at a fairly high level but uses important summary information such as relevant characteristics and capabilities to further guide the selection of a simulation tool based on project type, capabilities, and level of fidelity. A more detailed version of this information for each simulation tool is provided in Appendix C.

The reader is advised that this information represents simulation tool information current as of the publication of this report.

Airspace Analysis and Simulation Tools

There is no faster, cost-effective, and proven method to analyze new and revised airspace designs than through the fast-time modeling and simulation of airspace operations. The use of simulation tools to quantitatively support airspace operational changes can support capacity, growth, and safety decisions. Airspace simulations can relatively quickly and cheaply answer such questions as:

- How much will delays decrease with new procedures?
- Are the redesigned or new procedures sufficient to address the overall airspace operations and reduce potential hot spots or conflicts?
- How will future air traffic demand impact operations on the existing airport infrastructure?
- What is the best concept of operations to handle traffic to and from a given runway, and what added changes in the airspace may be required?

These airspace simulation tools can support and build on subject matter expertise and out-of-the-box spreadsheet models that often cannot sufficiently account for airspace complexity and dependencies. Table 20 lists commonly used airspace simulation tools as well as application and study scopes, sample outputs, and levels of customization. Generally, highly specialized simulation tools tend to have lower customization options compared to lower fidelity, more generic tools.

Airfield and Airport Analysis and Simulation Tools

There are considerable levels of uncertainty in different elements of airport and airfield operations that can affect the optimal performance of an airport. Several simulation tools—shown in Table 21—exist that can help airports understand the operational impacts of proposed airfield enhancements in terms of added safety, additional capacity, reduced runway occupancy time, or perhaps shorter taxi-in/taxi-out times.

Copyright National Academy of Sciences. All rights reserved.

Table 20. Airspace simulation tools.

Simulation Tool	Scope	Licensor	License Type	Product Support/Last Release
ADSIM	Runway, taxiway, apron	FAA	Public	Unknown
AirTOp	Airspace and airfield	Airtopsoft	Commercial	February 2018
CAST Aircraft	Airspace and airfield	Airport Research Center GmH	Commercial	Current
DART	Airspace (NAS), runway, weather	AvMet	Commercial	Current
FACET	Airspace (NAS), ATM concepts	NASA	Public	Unknown
FLAPS	Runway, arrival, and departure fixes	Flight Transportation Associates	Proprietary	Unknown
RAMS Plus	Airspace and airfield	RAMS Plus	Commercial	July 2016
SIMMOD Pro	Airspace and airfield	ATAC	Commercial	February 2017
systemwideModeler	NAS	MITRE	Proprietary	Unknown
TAAM	Airspace and airfield	Jappesen/Boeing	Commercial	February 2018
TARGETS	Procedure design	MITRE	Commercial	Current

Terminal Passenger Flow Analysis and Simulation Tools

Simulation tools today can process and analyze passenger behavior and flows in powerful 3-D environments, which greatly improves planning and visualization capabilities. Table 22 provides a list of some of the most commonly used simulation tools for terminal passenger flow planning.

Airport Terminal Systems Analysis and Simulation Tools

Airport terminal process simulations are typically built using specialized modules of more generic discrete-event or Monte Carlo simulation tools. This is primarily the case because simulations of terminal processes such as baggage systems, interterminal train links, and bus systems follow similar basic capacitated resource queueing model logic and analysis techniques. So, although the potential list of more generic simulation tools is virtually endless, the simulation tools listed in Table 23 are the models most referenced.

Curbside Vehicle Analysis and Simulation Tools

Curbside simulation tools were conceived to allow airport operators to evaluate the impact that landside processes have on terminal operations. This includes the analysis of parking lots, curbside operations, public transportation, taxi operation, airport access, exits, and through traffic, which could include traffic unrelated to passengers. Several simulation tools exist today that support the planning and analysis of curbside traffic and movements, which directly feed terminal design and passenger flow through the terminal. Table 24 presents a list of some of the most commonly used tools today in the planning and analysis of curbside operations.

Copyright National Academy of Sciences. All rights reserved.

Table 21. Airport and airfield simulation tools.

Simulation Tool	Scope	Licensor	License Type	Product Support/Last Release
AirTOp	Terminal control area, runway, taxiway, apron, gate	Airtopsoft	Commercial	February 2018
ArcPORT	Taxiway, apron, terminal	Transoft Solutions	Commercial	Current
AviPLAN Airside PRO	Taxiway, apron design	Transoft Solutions	Commercial	October 2017
CAST: Aircraft	Terminal control area, runway, taxiway, apron, gate	Airport Research Center GmbH	Commercial	July 2017
DART	Terminal control area and runways	AvMet	Commercial	Current
DELAYSIM	Runway	Flight Transportation Associates	Proprietary	Unknown
FLAPS	Runway, arrival, and departure fixes	Flight Transportation Associates	Proprietary	Unknown
RAMS Plus	Terminal control area, runway, taxiway, apron	RAMS Plus	Commercial	July 2016
RDSIM	Runway, runway exits	FAA	Public	Unknown
REDIM	Runway	VT Air Transportation Systems Laboratory	Public	Expected 2018
runwaySimulator	Runway	MITRE	Public	Current
SIMMOD Pro	Terminal control area, runway, taxiway, apron, gate	ATAC	Commercial	February 2017
TAAM	Terminal control area, runway, taxiway, apron, gate	Jeppesen/Boeing	Commercial	February 2018

Copyright National Academy of Sciences. All rights reserved.

Table 22. Terminal passenger simulation tools.

Simulation Tool	Scope	Licensor	License Type	Product Support/Last Release
AirTOp	Check-in, security, baggage claim, and customs	Airtopsoft	Commercial	February 2018
ArcPORT	Terminal area	Transoft Solutions	Commercial	Current
AutoMod	Baggage handling system, passenger flow	Brooks Automation/Simul8	Commercial	Current
CAST Terminal	Check-in counter, security control	Airport Research Center	Commercial	July 2017
FlexSim	Baggage handling system, passenger flow	FlexSim	Commercial	August 2017
MassMotion	Passenger terminal area	Oasys Software	Commercial	Current
PAX2SIM	Check-in counter, security control	Hub Performance	Proprietary	Current
PAXSIM	Check-in counter, baggage, security, boarding, customs	Jeppesen/Boeing	Commercial	Unknown
SimWalk Airport	Check-in counter, security control	SIMWALK	Commercial	Current

Table 23. Airport system simulation models.

Simulation Tool	Scope	Licensor	License Type	Product Support/Last Release
Arena	Terminal area	Arena Simulation	Commercial	September 2016
ExtendSim	Terminal area	ExtendSim	Commercial	November 2017
PTV Vissim	Terminal area	PTV Group	Commercial	2017
Simio	Terminal area	Simio	Commercial	July 2017

Table 24. Ground-access simulation models.

Simulation Tool	Scope	Licensor	License Type	Product Support/Last Release
CAST Curbside	Curbside, roadway, parking	Airport Research Center	Commercial	July 2017
PTV Vissim	Curbside, roadway	PTV Group	Commercial	October 2016
QATAR	Curbside, roadway	Transportation Research Board	Public	2010
Synchro	Traffic signal	Trafficware	Commercial	January 2017
TRACS	Terminal, roadway, curbside	Transsolutions	Proprietary	Unknown

Copyright National Academy of Sciences. All rights reserved.

CHAPTER 6

Conclusions and Future Research

Simulation tools clearly represent an integral part of the process for airport planning and design. More than 80 percent of the survey responders indicated that they use simulation tools to support studies ranging from major long-term master plans to day-to-day operational analyses. This clearly suggests that the simulation tool choice is an integral part of the decision process to ensure the quality and reliability of the overall study and analysis.

Findings

Nearly all airport planners that responded to the survey indicated that they believe simulation tools are an integral and necessary part of the airport planning process. However, the decision about which simulation tools to use and when and at which fidelity level to apply the tools should be made intelligently to maximize the overall study benefits and use time and resources efficiently.

As one of the survey responders indicated:

> Every airport and project is unique, and every application of every tool is different, therefore regardless of how simple or complex the tool is, the analysis will likely be time consuming and costly (there is normally no quick answer).

Keeping this mantra in mind, it is clear that some areas and scenarios are harder to simulate than others. As Figure 17 illustrates, airspace and airfield simulations are generally the most complex models to develop and analyze. This is primarily due to their high input data requirements, functional complexity, and the level of difficulty in setting up operationally valid scenarios. Medium levels of complexity and levels of effort are generally exhibited by airport and airfield simulation tools and by more generic passenger flow and terminal process simulations. Curbside simulation studies are generally less complex and resource intensive than those of other areas, at least from a simulation setup perspective.

However, it is not just the simulation tool that has an impact on simulation study complexity and level of effort. The amount of data processing, the extent of the experimental design, and the level of statistical analysis can all have significant impacts on the complexity of a simulation study.

Another point that was made by a survey responder is that "*Tools have no capabilities. The analyst determines the quality of the analysis.*" Hence, the application of a simulation tool to evaluate a specific operational or design problem is one thing. But unless the analyst understands the problem, knows how to apply the simulation tool correctly, and knows how to analyze data correctly, the results of a simulation study may very well be irrelevant and incorrect. Using unproven methodologies or varying from industry practices can create significant challenges in

Copyright National Academy of Sciences. All rights reserved.

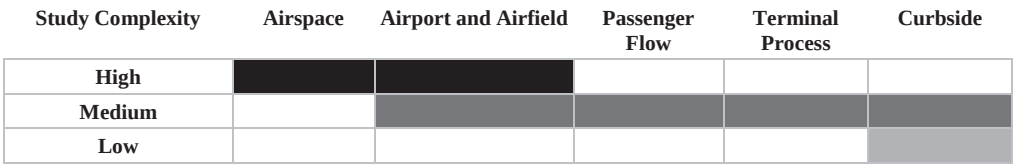


Figure 17. Survey results—simulation study level of complexity and effort.

trying to retain competence in the simulation tool and confidence in the analysis and results. Confidence needs to be gained by project stakeholders using sound postanalysis practices and validation steps.

Any simulation study should have quality assurance and control processes in place to ensure overall success. Processes should be in place to guide the development and execution of a simulation study using industry-accepted data standards, practices, and analyses methods. Other, more generic, engineering principles also need to be considered, including document and information configuration control and the verification and validation of simulation configurations and execution plans. A sound process that manages simulation input data across multiple scenarios and configurations is necessary, particularly for larger and more complex simulation studies.

For many projects, a simulation tool should also be considered part of a suite of potential tools. Often, ensembles of simulation tools need to be used to produce complete and reliable results. This is particularly true when novel concepts are being simulated or when interfaces between different modes (passengers and aircraft or vehicles and passengers) need to be considered.

As an organization, airports need to understand and accept when it is better to outsource simulation studies to consultants that are more familiar with specific tools and analysis practices. This decision needs to be based on an evaluation of the availability of internal knowledgeable resources and the potential cost and impact of the study.

Capability Gaps and Problem Areas

More than 50 percent of the survey responders felt satisfied with current simulation tool capabilities. This response rate may partly be a result of expectation management on the analyst's side. No simulation tool will ever provide 100 percent of the desired functionality, but a seasoned analyst understands these limitations and knows when and how to use specific tools to answer specific questions.

Several basic issues and problem areas with current simulation tools and their application were identified in the survey. The following rank-ordered list provides some insight into areas of perceived simulation study risk as well as missing capabilities in current tools:

1. Amount of experience required to gain proficiency (i.e., learning curve).
2. Usability, ease of use (e.g., user interface, software interaction, setup).
3. Effort required to format and ability to import data.
4. Ability to generate output compatible with follow-on analyses.
5. Quality of visualization (e.g., graphics, 2-D versus 3-D, display customization, user interaction).
6. Ability to debug and troubleshoot models.
7. Ease of setting up and generating visualization.
8. Ability to customize output formats and content.
9. Lack of variables and guidance on human-factors-related assumptions (e.g., buffers, sensitivity tests).

Copyright National Academy of Sciences. All rights reserved.

Overall, survey responders indicated that most simulation tools lack quality visualization capabilities and require significant effort in terms of input data processing and output data analysis.

It is evident that the required experience level is a major decision factor, particularly when selecting higher fidelity simulation tools. Analysts need to understand when the use of simulation tools is appropriate for a study and what the basic study expectations are. A misunderstanding in these areas across the simulation team and external stakeholders can create significant issues.

Future Research

Although novel approaches and heuristics to airport and airspace planning and design, passenger flow analysis, and process optimization are continually introduced, the core methodologies and analysis concepts remain largely unchanged. Nevertheless, as software technology and information processing theories evolve, other simulation tools and techniques may require consideration.

The information on the use of simulation tools presented in this report represents the current state of industry practices in 2018. Ideally, as simulation tool functionality evolves and new tools enter the market, this information would be updated periodically. Such updates should also consider changes in regional preferences in simulation tools across the globe.

Also, as system complexity grows, parallel processing simulation software may support ever growing systems. Some tools such as AirTOp already support rudimentary parallel processing concepts, but the division of processing tasks and interactions between parallel simulations remains a significant research topic. Another simulation concept that will require consideration is agent-based modeling and simulation. In complex sociotechnical environments where systems, humans, and procedures interact, agent-based tools show promise in simulating emergent behavioral patterns with competing goals (Bouarfa et al., 2013).

The concept of safety may also require more future consideration. Although simulation metrics today are primarily geared around efficiency and cost, there is often a safety aspect associated with changes in traffic demand, procedures, or designs. Very few current simulation tools really address this safety concept or other related human factors metrics.

Copyright National Academy of Sciences. All rights reserved.

References

Barnhart, C., P. Belobaba, P., and A. R. Odoni. Applications of Operations Research in the Air Transport Industry. *Transportation Science*, Vol. 37, Issue 4, Nov. 2003, pp. 368–391.

Boesel, J., C. Gladstone, J. Hoffman, P. Massimini, C. Shiotsuki, and B. Simmons. *TAAM Best Practices Guideline*. MITRE Technical Report 01W0000092. McLean, Va., Sept. 2001.

Bouarfa, S., H. Blom, R. Curran, and M. Everdij. Agent-Based Modeling and Simulation of Emergent Behavior in Air Transportation., *Complex Adaptive Systems Modeling*. 1:15, Aug. 2013, pp. 1–26.

DFS Deutsche Flugsicherung GmbH. *Fast-Time Simulation Report–DFS Reference 2014–228*. DFS, Langen, Germany, Jan. 2015.

Duncan, G., and H. Johnson. Development and Application of a Dynamic Simulation Model for Airport Curbsides. In *The 2020 Vision of Air Transportation: Emerging Issues and Innovative Solutions*, S.S. Nambisan, American Society of Civil Engineers, San Francisco, Cal., 2000, pp. 153–164.

FAA. *Chicago O'Hare International Airport–Final EIS.* FAA, Washington, D.C., July 2005.

FAA. *Record of Decision for Capacity Enhancement Program at Philadelphia International Airport*. FAA, New York, Dec. 2010.

FAA. *FAA Regulations and Policies*. FAA, Washington, D.C., Oct. 2017. https://www.faa.gov/regulations_policies/policy_guidance/benefit_cost/. Accessed Oct. 25, 2017.

Fayez, M., A. Kaylani, D. Cope, N. Rychlik, and M. Mollaghasemi. Managing Airport Operations Using Simulation. *Journal of Simulation*. Vol. 2, Issue 1, 2008, pp. 41–52.

Gosling, G. D. Airport Landside Planning Techniques, Introduction. *Transportation Research Record: Journal of the Transportation Research Board*, No. 1199, 1988.

Guizzi, G., T. Murino, and E. Romano. A Discrete Event Simulation to Model Passenger Flow in the Airport Terminal. *Mathematical Methods and Applied Computing–Volume II*. WSEAS Press, Athens, Greece, 2009.

Harris, T. M., M. Nourinejad, and M. J. Roorda. A Mesoscopic Simulation Model for Airport Curbside Management. *Journal of Advanced Transportation*, Vol. 2017, Article ID 4950425, July 2017, pp. 1–19.

Kuzminski, P. C. An Improved runwaySimulator–Simulation for Runway System Capacity Estimation. Presented at Integrated Communications, Navigation and Surveillance Conference (ICNS), Herndon, Va., 2013.

Landrum & Brown et al. *ACRP Report 25*: *Airport Passenger Terminal Planning and Design*, *Volume 1: Guidebook.* Transportation Research Board of the National Academies, Washington, D.C., 2010.

LeighFisher; Dowling Associates, Inc.; and JD Franz Research, Inc. *ACRP Report 40: Airport Curbside and Terminal Area Roadway Operations*. Transportation Research Board of the National Academies, Washington, D.C., 2010.

LeighFisher; Landrum & Brown; CDM Smith; George Mason University; University of California, Berkeley; and Presentation & Design, Inc. *ACRP Report 79*: *Evaluating Airfield Capacity*. Transportation Research Board of the National Academies, Washington, D.C., 2012.

Manley, M., Y. Kim, K. Christensen, and A. Chen. Modeling Emergency Evacuation of Individuals with Disabilities. *Journal of the Transportation Research Board*. Vol. 2206, 2011, pp. 32–38.

Nelson, B. L. Some Tactical Problems in Digital Simulation for the next 10 years. *Journal of Simulation*, Vol. 10, No. 1, 2016, pp. 2–11.

Savrasovs, M., A. Medvedev, and E. Sincova. Riga Airport Baggage Handling System Simulation. Presented at 23rd European Conference on Modeling and Simulation, Madrid, Spain, 2009.

Sillard, L., F. Vergne, and B. Desart. *TAAM Operational Evaluation*. Eurocontrol Experimental Center Report No. 351–Project SIM-S-E8, Bretigny, France, Aug. 2000.

Copyright National Academy of Sciences. All rights reserved.

Solak, S., J.-P. B. Clarke, and E. L. Johnson. Airport Terminal Capacity Planning. *Transportation Research Part B: Methodological*, Vol. 43, Issue 6, July 2009, pp. 659–676.

TransSolutions; Futterman Consulting; Harris Miller Miller and Hanson, Inc; and Jasenka Rakas. *ACRP Report 104*: *Defining and Measuring Aircraft Delay and Airport Capacity Thresholds*. Transportation Research Board of the National Academies, Washington, D.C., 2014.

Wilson, I., and F. Hafner. Benefit Assessment of Using Continuous Descent Approaches at Atlanta. Presentation at 24th Digital Avionics and Systems Conference (DASC), Washington, D.C., 2005.

Copyright National Academy of Sciences. All rights reserved.

Abbreviations and Acronyms

ADSIM	Airport Delay Simulation Model
AMAN	Arrival Manager
ATC	Air Traffic Control
CAST	Comprehensive Airport Simulation Technology
CBA	Cost–Benefit Analysis
DART	Data Analysis Reporting Tool
DDFS	Design-Day Flight Schedule
DMAN	Departure Manager
FAA	Federal Aviation Administration
FACET	Future Airborne Communications Equipment and Technology
FLAPS	Flexible Airport Simulation
LOS	Level of Service
MDW	Chicago Midway International Airport
NASA	National Aeronautics and Space Administration
PAX	Passengers
PBN	Performance Based Navigation
QATAR	Quick Analysis Tool for Airport Roadways
RDSIM	Runway Delay Simulation
REDIM	Runway Exit Design Interactive Model
RNAV	Area Navigation
RNP	Required Navigation Performance
SESAR	Single European Sky ATM Research
SID	Standard Instrument Departure
SIMMOD	Airport and Airspace Simulation Model
STAR	Standard Terminal Arrival Route
SWAC	Systemwide Analysis Capability
TAAM	Total Airspace and Airport Modeler
TARGETS	Terminal Area Route Generation and Traffic Simulation
TFM	Traffic Flow Management
TMA	Traffic Management Advisor
TRACON	Terminal Radar Approach Control
TRACS	Terminal, Roadway, and Curbside Simulation

Copyright National Academy of Sciences. All rights reserved.

APPENDIX A

Survey Questionnaire

Contact Information

Please provide some basic information about yourself:*
First Name*: _____
Last Name*: _____
Title*: _____
Airport/Organization*: _____
Email Address*: _____
Phone Number: _____

Basic Information

1) Does your airport have a dedicated planning/design department?*
( ) Yes
( ) No

2) What is your role within your organization from an airport planning/design perspective?*
[ ] Airport Management
[ ] Airport Planning Department Manager/Director
[ ] Airport Planner
[ ] Simulation/Modeler
[ ] Airport Consultant
[ ] Other – Please comment:

3) How many airports does your organization manage or provide planning/design services for?*
( ) 0
( ) 1
( ) 2-3
( ) More than 3 – Please specify total #:

*Mandatory question.

Copyright National Academy of Sciences. All rights reserved.

52 Simulation Options for Airport Planning

4) The largest airport you conduct planning for is considered a:
( ) Large Hub
( ) Medium Hub
( ) Small Hub
( ) Non-Hub Primary
( ) Other – Please specify:
( ) None

5) Does your organization currently use simulation and modeling software tools as part of your airport planning process?*
( ) Yes
( ) No
( ) I am not sure – Please comment:

6) If you currently do not use simulation tools within your organization, please provide a reason why not: (check all that apply)
[ ] No need
[ ] Insufficient resources in-house
[ ] Cost of simulation tool/study
[ ] Desire to outsource to consultants
[ ] Other – Please provide comment:

7) May we contact you for follow-up questions regarding your responses?
( ) Yes
( ) No

Overall Simulation and Modeling Usage

8) For which operational areas have you or your organization used simulation and modeling software tools? (check all that apply)
*Note: Your questionnaire will be tailored based on the areas you select**
[ ] Airfield/Airside – (runways, aprons, gates, aircrafts, ground service equipment, etc.)
[ ] Airspace – (TMA/TRACON airspace, airspace procedure design/evaluation, etc.)
[ ] Terminal – (passenger movements, baggage systems, security screening areas, check-in queues, etc.)
[ ] Ground Access – (curb-side processes, passenger drop-off/pickup areas, road access, etc.)
[ ] Other – Please provide Comment:
[ ] None

9) For what purposes do you typically use simulation tools? (check all that apply)
[ ] Master Planning
[ ] Capacity/Demand Planning
[ ] Airport Design
[ ] Operations Planning
[ ] Procedure/Process Design
[ ] Contingency/Irregular Operations Planning
[ ] Security/Safety Studies
[ ] Other – Please provide comment:

Copyright National Academy of Sciences. All rights reserved.

10) Would you consider your use of simulation tools to be tactical (day-to-day operations/capacity planning) or strategic (master planning, capital improvements, procedure design, etc.)? (check all that apply)
[ ] Tactical – Please provide comments:
[ ] Strategic – Please provide comments:

11) Do you have dedicated/trained staff to create and/or operate simulation software tools for airport planning/design?*

( ) Yes
( ) No

12) Do you outsource modeling and simulation tasks to outside consultants?*
( ) Yes
( ) No

13) Which secondary models do you use that are fed by simulation tool results/output? (check all that apply)
[ ] Environmental/Noise/Emissions
[ ] Procedure Development Tools
[ ] Cost/Benefit or Financial Analysis Tools
[ ] Other – Please provide comment:
[ ] None

14) What factors do you consider in your selection of simulation tools for a planning/design study? (check all that apply)
[ ] Cost/Schedule/Resources
[ ] Project Type
[ ] Data Availability and Input Requirements
[ ] Simulation Output/Results and Presentation/Visualization Capabilities
[ ] Simulation Tool Fidelity (modeling/result precision, reliability of results, etc.)
[ ] Industry/Government Acceptance of Simulation Tool (Track Record)
[ ] Other – Please provide comment:

Airfield/Airside Simulation Tools

15) How frequently does your organization conduct airfield simulation studies?*
( ) Never
( ) Infrequently
( ) Once or twice per year
( ) More than 2 times per year
( ) On-Going effort
( ) Other – Please provide comment:

Copyright National Academy of Sciences. All rights reserved.

54 Simulation Options for Airport Planning

16) Which airfield simulation and modeling tools have been used by your organization? (check all that apply)*
[ ] Purpose-built Commercial – Please provide names:
[ ] Generic Commercial (i.e., Excel) – Please provide names:
[ ] In-House – Please provide description:

17) If you do use commercial purpose-built airfield simulation tools, do you own licenses to the simulation software?
( ) Yes
( ) No, we rent/lease when needed

18) What is the general purpose for conducting airfield simulation studies for your organization? (check all that apply)
[ ] Runway/Gate/Airfield Capacity Planning and Design (Master Planning, etc.)
[ ] Construction/Operations Planning
[ ] Schedule/Gate/Resource Planning
[ ] Irregular Operations Planning (De-Ice Scenarios, etc.)
[ ] Other – Please provide comment:

19) Which secondary post-processing applications do you use that are fed by outputs/results from your airfield simulation tools? (check all that apply)
[ ] Environmental/Noise/Emissions
[ ] Cost/Benefit or Financial Analysis Tools
[ ] Visualization/Presentation Tools
[ ] Custom Simulation Reporting and Analysis Tools
[ ] Other – Please provide comment:
[ ] None

20) What is your perceived level of difficulty for developing and analyzing airfield simulation projects within your organization?
( ) Low – Input data is easily obtained, scenarios are easily developed and analyzed
( ) Medium – Input data takes effort to process, scenarios development takes some time and results are not easy to extract and interpret
( ) High – Input data is hard to obtain or process, development of scenarios requires significant resources and expertise, analysis of results takes expertise and in some cases requires custom post-processing tools
( ) Other – Please provide comment:

Airspace Simulation Tools

21) How frequently does your organization conduct airspace simulation studies?*
( ) Never
( ) Infrequently
( ) Once or twice per year
( ) More than 2 times per year
( ) On-Going effort
( ) Other – Please provide comment:

Copyright National Academy of Sciences. All rights reserved.

22) Which airspace simulation and modeling tools have been used by your organization? (check all that apply)*
[ ] Purpose-built Commercial – Please provide names:
[ ] Generic Commercial (i.e., Excel) – Please provide names:
[ ] In-House – Please provide description:

23) If you do use commercial purpose-built airspace simulation tools within your organization, do you own licenses to the simulation software?
( ) Yes
( ) No, we rent/lease when needed

24) What is the general purpose for conducting airspace simulation studies within your organization?
[ ] Runway/Airport Capacity Planning
[ ] TMA/TRACON Capacity Planning
[ ] Procedure Design/Evaluation
[ ] Evaluation TMA/TRACON impacts on Airport Operations
[ ] Other – Please provide comment:

25) Which secondary post-processing applications do you use that are fed by outputs/results from your airspace simulation tools?
[ ] Environmental/Noise/Emissions
[ ] Cost/Benefit or Financial Analysis Tools
[ ] Visualization/Presentation Tools
[ ] Custom Simulation Reporting and Analysis Tools
[ ] Other – Please provide comment:
[ ] None

26) What is your perceived level of difficulty for developing and analyzing airspace simulation projects within your organization?
[ ] Low – Input data is easily obtained, scenarios are easily developed and analyzed
[ ] Medium – Input data takes effort to process, scenarios development takes some time and results are not easy to extract and interpret
[ ] High – Input data is hard to obtain or process, development of scenarios requires significant resources and expertise, analysis of results takes expertise and in some cases requires custom post-processing tools
[ ] Other – Please provide comment:

Terminal Simulation Tools

27) How frequently does your organization conduct terminal simulation studies (e.g. passengers, bags, staffing, logistics, etc.)?*
( ) Never
( ) Infrequently
( ) Once or twice per year
( ) More than 2 times per year
( ) On-Going effort
( ) Other – Please provide comment:

Copyright National Academy of Sciences. All rights reserved.

56 Simulation Options for Airport Planning

28) Which terminal simulation and modeling tools have been used by your organization? (check all that apply)*
[ ] Purpose-built Commercial – Please provide names:
[ ] Generic Commercial (i.e., Excel) – Please provide names:
[ ] In-House – Please provide description:

29) If you do use commercial purpose-built terminal simulation tools within your organization, do you own licenses to the simulation software?
( ) Yes
( ) No, we rent/lease when needed

30) What is the general purpose for conducting terminal simulation studies within your organization?
[ ] Terminal Passenger Capacity Planning
[ ] Baggage System and Operations Planning
[ ] Passenger/Terminal Process Design/Planning
[ ] Security/Check-in Process and Capacity Planning
[ ] Staff/Resource Planning
[ ] Other – Please provide comment:

31) Which secondary post-processing applications do you use that are fed by outputs/results from your terminal simulation tools?
[ ] Cost/Benefit or Financial Analysis Tools
[ ] Visualization/Presentation Tools
[ ] Custom Simulation Reporting and Analysis Tools
[ ] Other – Please provide comment:
[ ] None

32) What is your perceived level of difficulty for developing and analyzing terminal simulation projects within your organization?
[ ] Low – Input data is easily obtained, scenarios are easily developed and analyzed
[ ] Medium – Input data takes effort to process, scenarios development takes some time and results are not easy to extract and interpret
[ ] High – Input data is hard to obtain or process, development of scenarios requires significant resources and expertise, analysis of results takes expertise and in some cases requires custom post-processing tools
[ ] Other – Please provide comment:

Ground-Access Simulation Tools

33) How frequently does your organization conduct ground-access simulation studies?*
( ) Never
( ) Infrequently
( ) Once or twice per year
( ) More than 2 times per year
( ) On-Going effort
( ) Other – Please provide comment:

Copyright National Academy of Sciences. All rights reserved.

34) Which ground-access simulation and modeling tools have been used by your organization? (check all that apply)*
[ ] Purpose-built Commercial – Please provide names:
[ ] Generic Commercial (i.e., Excel) – Please provide names:
[ ] In-House – Please provide description:

35) If you do use commercial purpose-built ground-access simulation tools within your organization, do you own licenses to the simulation software?
( ) Yes
( ) No, we rent/lease when needed

36) What is the general purpose for conducting ground-access simulation studies within your organization? (check all that apply)
[ ] Airport Terminal Access Planning
[ ] Airport Road Access Planning
[ ] Other – Please provide comment:

37) Which secondary post-processing applications do you use that are fed by outputs/results from your ground-access simulation tools? (check all that apply)
[ ] Cost/Benefit or Financial Analysis Tools
[ ] Visualization/Presentation Tools
[ ] Custom Simulation Reporting and Analysis Tools
[ ] Other – Please provide comment:
[ ] None

38) What is your perceived level of difficulty for developing and analyzing ground-access simulation projects within your organization?
( ) Low – Input data is easily obtained, scenarios are easily developed and analyzed
( ) Medium – Input data takes effort to process, scenarios development takes some time and results are not easy to extract and interpret
( ) High – Input data is hard to obtain or process, development of scenarios requires significant resources and expertise, analysis of results takes expertise and in some cases requires custom post-processing tools
( ) Other – Please provide comment:

Simulation Results and Analysis

39) What simulation outputs do you frequently use for reports, presentations, and discussion? (check all that apply)
[ ] Detailed simulation metrics
[ ] Derived Excel-based tables/graphs
[ ] Visualizations of simulations
[ ] Others – Please comment:

40) Do you use simulation data to substantiate benefit/cost analyses or capital improvement plans?
( ) Yes
( ) No – Please comment why not:

Copyright National Academy of Sciences. All rights reserved.

41) Do you find simulation results to be generally reliable and truthful in forecasting operational issues?
( ) Yes
( ) No – Please comment why not:

42) What percentage of the overall simulation and modeling effort do you generally feel consists of initially calibrating models to historical events/data and on-going maintenance to keep models current?
( ) Minimal/Negligible
( ) less than 10%
( ) 10%-25%
( ) 25%-50%
( ) more than 50%

43) What types of statistical analyses do you typically use to post-process/analyze/compare simulation metrics and results? (check all that apply)
[ ] Means/Averages (hourly, daily, by aircraft/passenger, etc.)
[ ] Histograms/Bar Charts
[ ] Percentiles or Confidence Intervals (i.e., 95%)
[ ] Other – Please describe:

44) Do you generally use multi-runs within your simulation studies to add randomness or increase statistical validity of results?
( ) Yes – always
( ) Yes – sometimes
( ) No – please comment on reason (if any):
( ) I don't know

Benefits, Functionality, Capability Gaps, etc.

45) What benefits do you see in using simulation and modeling tools to support airport planning/design? (check all that apply)
[ ] Proof of concept, decision support
[ ] Generating data to support legally-required analysis (e.g., environmental, financial, etc.)
[ ] Generating data to support design
[ ] Predicting the future, near-term (daily planning)
[ ] Predicting the future, long-term (long-range planning)
[ ] Other – Please provide comment:

46) Are you currently satisfied with the functionality that simulation software tools provide to support airport planning/design?
( ) Yes
( ) No – Please provide comment on why not:
( ) Not enough background to say

Copyright National Academy of Sciences. All rights reserved.

47) What functionality do you find missing or feel requires improvement in current simulation and modeling tools to improve usability for airport planning/design? (check all that apply)
[ ] Usability, ease of use (e.g., user interface, software interaction, set-up)
[ ] Amount of experience required to gain proficiency (i.e., learning curve)
[ ] Effort required to format and ability to import data
[ ] Ability to debug and troubleshoot models
[ ] Ability to generate output compatible with follow-on analyses
[ ] Ability to customize output formats and content
[ ] Ease of setting up and generating visualization
[ ] Quality of visualization (e.g., graphics, 2D vs. 3D, display customization, user interaction, etc.)
[ ] Other – Please provide comment:

48) If you or your organization have used simulation tools for airport planning, do you generally find that software companies provide timely updates and support if there are any issues?
( ) Yes
( ) No – Please provide comments:
( ) I don't know

Additional Information

49) What specific information regarding simulation and modeling tools and their application to airport planning/design do you think would be valuable to airport managers/planners in a summary report?

50) Is there any other information you would like to share regarding your personal/company use or experience with simulation and modeling tools for airport planning and design that may not have been covered in this survey?

Copyright National Academy of Sciences. All rights reserved.

APPENDIX B

Sample Simulation Study Inputs and Outputs

Airspace Studies

Typical Inputs	Aircraft Performance Profiles Flight Schedule Airport/Runway Layouts Runway Dependency and Coordination SIDs and STARs Sector Maps
Typical Outputs	Airport/Runway Throughput and Capacity Airspace/Fix Throughput and Capacity Airspace/Airport/Runway Delay Potential Conflict Counts and Areas Sector and Fix Usage Travel Time and Distance Fuel Burn Noise/Emission Plots

Airfield Studies

Typical Inputs	Aircraft Performance Profiles Flight Schedule Airport/Runway Layouts Runway Dependency and Coordination SIDs and STARs Gate and Apron Usage and Coordination Gate Allocation Plans Taxiway Usage and Restrictions
Typical Outputs	Airport/Runway Throughput and Capacity Airport/Runway Delay Travel Time and Distance Fuel Burn Taxi-Flow Bottlenecks Runway Queue Lengths Runway Occupancy Time and Exit Locations

Copyright National Academy of Sciences. All rights reserved.

Terminal Passenger Studies

Typical Inputs	Statistical Distributions of Passenger Characteristics, Behavior, and Trends Origin-Destination Versus Connecting Passenger Information Passenger Arrival Statistics and Trends (i.e., Early Arrivals) Flight Schedules Detailed Terminal Layouts Passenger Flow Maps Terminal Resource Maps and Characteristics (check-in kiosks, security checkpoints, etc.)
Typical Outputs	Passenger Flow Maps Resource Utilization, Capacity, and Throughput Queue Lengths Flow Bottleneck Passenger Transit Times Wait Times and Delays

Terminal System Studies

Typical Inputs	Passenger Demand Statistics Passenger Characteristics and Demographics Flight Schedules System Resource Operational Statistics System Resource Time Schedules System Process Time Statistics System Design Layouts
Typical Outputs	System Capacity System Throughput Resource Utilization Delays and Queue Lengths Bottlenecks

Ground-Access Studies

Typical Inputs	Passenger Arrival and Departure Statistics Flight Schedules Vehicle Statistics Detailed Ground-Access or Roadway Layouts Passenger Crossings
Typical Outputs	Capacity Throughput Utilization of Ground-Access Service Resources Vehicle Delays and Queue Lengths Bottlenecks

Copyright National Academy of Sciences. All rights reserved.

APPENDIX C

Simulation Model Characteristics

ADSIM

Scope	Airspace and airside
Fidelity	Medium
Functionality and Capabilities	Movement of aircraft on the airport surface and in the immediate airspace, composed of the common approach and departure corridors.
Applicable Project	Runway capacity planning
Simulation Output	Hourly arrival and departure flow rates, travel time, arrival and departure delays.
Visualization Capabilities	Yes

AirTOp

Scope	Airspace, airside, and terminal
Fidelity	High
Functionality and Capabilities	En-route traffic and ATC modeling Advanced Airport TMA/TRACON modeling Network and Flow Management modeling (taxiway, terminal)
Applicable Project	Airspace planning, airside planning, terminal planning
Simulation Output	Capacity, delay, conflict, fuel burn, processing time
Visualization Capabilities	Yes

Copyright National Academy of Sciences. All rights reserved.

ArcPort

Scope	Taxiway, apron, terminal
Fidelity	High
Functionality and Capabilities	Analyze and visualize flows of passengers, aircraft, vehicles, baggage, and cargo within the airport infrastructure, in fast time.
Applicable Project	Airside planning, terminal planning, landside planning
Simulation Output	Capacity, delay
Visualization Capabilities	Yes

Arena

Scope	Generic tool
Fidelity	Medium–High
Functionality and Capabilities	Build, analyze, and visualize flows of passengers, vehicles, baggage, and cargo within the airport infrastructure.
Applicable Project	Airside planning, terminal planning, landside planning
Simulation Output	Detailed statistics
Visualization Capabilities	Yes

AutoMod

Scope	Generic tool
Fidelity	Medium–High
Functionality and Capabilities	Build, analyze, and visualize flows of passengers, vehicles, baggage, and cargo within the airport infrastructure.
Applicable Project	Airside planning, terminal planning, landside planning
Simulation Output	Detailed statistics
Visualization Capabilities	Yes

AviPlan Airside Pro

Scope	Taxiway, apron
Fidelity	Medium
Functionality and Capabilities	Aircraft and ground vehicle simulations and accounting. Aircraft pushback operations both without towbars and with tractor and towbar configurations. Jet blast contours and impact surfaces for various engine thrust levels.
Applicable Project	Airside planning, design, and operations
Simulation Output	Jet blast contour, nose gear, or cockpit ground marking in Word and/or CAD format.
Visualization Capabilities	Yes

Copyright National Academy of Sciences. All rights reserved.

Comprehensive Airport Simulation Technology (CAST): Aircraft

Scope	Runway, taxiway
Fidelity	High
Functionality and Capabilities	Multiagent fast time simulation of aircraft traffic schedule-based traffic generation. Dynamic aircraft stand allocation. ATMrunway and traffic management. Sophisticated routing and taxiing behavior.
Applicable Project	Airspace planning, airside planning, terminal planning
Simulation Output	Taxi times, average waiting times, queue length, flow rates, throughput, delays, number of runway crossings, and runway occupancy times.
Visualization Capabilities	Yes

Comprehensive Airport Simulation Technology (CAST): Curbside

Scope	Curbside
Fidelity	High
Functionality and Capabilities	Flexible and efficient road network set-up Several options for automated traffic demand generation Demand generation based on flight schedule or “random” traffic generation. Scalable level of detail: quick facility requirement analysis or detailed simulation.
Applicable Project	Landside planning and management
Simulation Output	Travel and process times, parking and road utilization, delays
Visualization Capabilities	Yes

Comprehensive Airport Simulation Technology (CAST): Terminal

Scope	Terminal
Fidelity	High
Functionality and Capabilities	Evaluation of terminal layouts and operational strategies, identification of potential in capacity, passenger flow simulation.
Applicable Project	Terminal planning
Simulation Output	Utilization of stopover and movement areas as well as waiting queue development. Identification of bottlenecks.
Visualization Capabilities	Yes

Copyright National Academy of Sciences. All rights reserved.

DART

Scope	NAS, Air Traffic Management
Fidelity	Medium
Functionality and Capabilities	Integrated system-wide airport, airspace, weather, and TMI simulation.
Applicable Project	Air Traffic Management and Concept Evaluation
Simulation Output	Delays, cancellations, diversions, excess miles flown, fuel burn, airport movement, sector demand and capacity, sector occupancy, individual flight statistics
Visualization Capabilities	Yes

DELAYSIM

Scope	Runway
Fidelity	Medium
Functionality and Capabilities	Simulate how air traffic controllers might use the airport's runways based on specified demand characteristics and actual weather observations. Operates by sequencing through hourly weather observations and simulating the controllers' selection of runway for each hour.
Applicable Project	Runway capacity planning
Simulation Output	Runway capacity and detailed statistics on delays, runway use, and exit utilization by aircraft class
Visualization Capabilities	No

ExtendSim

Scope	Generic tool
Fidelity	Medium–High
Functionality and Capabilities	Build, analyze, and visualize flows of passengers, vehicles, baggage, and cargo within the airport infrastructure.
Applicable Project	Airside planning, terminal planning, landside planning
Simulation Output	Detailed statistics
Visualization Capabilities	Yes

Copyright National Academy of Sciences. All rights reserved.

Future ATM Concepts Evaluation Tool (FACET)

Scope	Air traffic management
Fidelity	Medium
Functionality and Capabilities	Advanced ATM concept development and evaluation
Applicable Project	Airspace planning, concept evaluation
Simulation Output	Sector, airport, fix, and airway usage
Visualization Capabilities	Yes

Flexible Airport Simulation (FLAPS)

Scope	Runway
Fidelity	Medium
Functionality and Capabilities	Modeling aircraft from the approach fix to the runway exit and runway departure queue to the departure fix
Applicable Project	Runway capacity planning
Simulation Output	Detailed statistical outputs on runway capacity and utilization, aircraft delays, exit use, and runway queues
Visualization Capabilities	No

FlexSim

Scope	Generic tool
Fidelity	Medium–High
Functionality and Capabilities	Build, analyze, and visualize flows of passengers, vehicles, baggage, and cargo within the airport infrastructure.
Applicable Project	Airside planning, terminal planning, landside planning
Simulation Output	Detailed statistics
Visualization Capabilities	Yes

MassMotion

Scope	Terminal area (pedestrian and crowd)
Fidelity	Medium–High
Functionality and Capabilities	Pedestrian simulation and crown analysis Capable of simulating hundreds of thousands of people within a matter of hours
Applicable Project	Terminal planning, landside planning
Simulation Output	Detailed statistics
Visualization Capabilities	Yes

Copyright National Academy of Sciences. All rights reserved.

Pax2Sim

Scope	Passenger terminal
Fidelity	Medium
Functionality and Capabilities	Data-driven solution to plan resources and optimize passenger flows and waiting times in terminals.
Applicable Project	Resource planning, passenger flow management
Simulation Output	Queue occupation stats, throughput
Visualization Capabilities	Yes

PaxSim

Scope	Terminal
Fidelity	Medium
Functionality and Capabilities	Passenger flows, check-in and carry-on baggage processing and flows
Applicable Project	Terminal planning
Simulation Output	Detailed stats on passengers and flows
Visualization Capabilities	Yes

Quick Analysis Tool for Airport Roadways (QATAR)

Scope	Roadway
Fidelity	Low
Functionality and Capabilities	Interaction between curbside and bypass lanes
Applicable Project	Landside planning and management
Simulation Output	Delay, throughput, utilization
Visualization Capabilities	No

RAMS Plus

Scope	Airspace, Air Traffic Control
Fidelity	High
Functionality and Capabilities	Flight trajectory, spatial conflict detection and rule-based conflict resolution, TMA, time-based metering with user-defined subscription, runway scheduling and runway usage conflicts, weather and wind.
Applicable Project	Airside planning
Simulation Output	Workload, capacity, delay
Visualization Capabilities	Yes

Copyright National Academy of Sciences. All rights reserved.

RDSIM (Runway Delay SIM)

Scope	Runway
Fidelity	Medium
Functionality and Capabilities	Trace the path of each aircraft through space and time on the runway network. Consider implicitly only the final approach and initial departure airspace.
Applicable Project	Runway capacity and planning
Simulation Output	Hourly airport capacity Total and average aircraft delays
Visualization Capabilities	No

REDIM

Scope	Runway
Fidelity	Medium
Functionality and Capabilities	Simulate runway roll from threshold to runway exit
Applicable Project	Runway exit design and valuation
Simulation Output	Runway exit utilization, occupancy time
Visualization Capabilities	No

runwaySimulator

Scope	Runway
Fidelity	Medium
Functionality and Capabilities	Simulate arriving and departing traffic taking into account the decisions made about runway assignment and sequencing.
Applicable Project	Runway capacity planning
Simulation Output	Capacity, maximum runway throughput
Visualization Capabilities	Yes

Copyright National Academy of Sciences. All rights reserved.

Simio

Scope	Generic tool
Fidelity	Medium–High
Functionality and Capabilities	Build, analyze, and visualize flows of passengers, vehicles, baggage, and cargo within the airport infrastructure.
Applicable Project	Airside planning, terminal planning, landside planning
Simulation Output	Detailed statistics
Visualization Capabilities	Yes

SIMMOD Pro

Scope	Runway, taxiway, apron
Fidelity	High
Functionality and Capabilities	Capable of simulating full individual airfield (runway, taxiway, and apron), an airfield and its associated terminal airspace, regional system of airports and its associated terminal airspace.
Applicable Project	Airspace planning, airside planning
Simulation Output	Travel times, traffic flows past specified points, throughput capacity, delay, fuel burn
Visualization Capabilities	Yes

SIMWALK Airport

Scope	Terminal area
Fidelity	High
Functionality and Capabilities	Integrate all standard airport facilities (check-in counter, security, immigration). Airport evacuation simulation.
Applicable Project	Terminal planning, crowd and people flow management
Simulation Output	Facility capacity and delay, queue length
Visualization Capabilities	Yes

Copyright National Academy of Sciences. All rights reserved.

70 Simulation Options for Airport Planning

Synchro

Scope	Ground traffic analysis and optimization
Fidelity	High
Functionality and Capabilities	Design, model, and optimize traffic signal infrastructure
Applicable Project	Ground traffic analysis and management
Simulation Output	Capacity, throughput, delay
Visualization Capabilities	Yes

systemwideModeler

Scope	NAS System
Fidelity	Medium
Functionality and Capabilities	NAS-wide simulation tool. Capable of measuring impacts on traffic loads (airport, arrival/departure queues), sectors, airways, controller workload, fuel burn.
Applicable Project	Air traffic management, system design and concept evaluation
Simulation Output	Delay, workload, fuel burn, sector utilization
Visualization Capabilities	Yes but postprocessing required

TARGETS

Scope	Procedure (Arrival/Departure) design
Fidelity	Medium
Functionality and Capabilities	Design, analysis, and operational assessments of procedures and airspace
Applicable Project	Procedure design
Simulation Output	Flyability (FAA compliance)
Visualization Capabilities	Yes

Copyright National Academy of Sciences. All rights reserved.

TRACS (Terminal, Roadway, and Curbside Simulation)

Scope	Roadway, curbside
Fidelity	High
Functionality and Capabilities	Driver behavior, multimodal transit operations, interface with planning and forecasting models, and 3-D simulation.
Applicable Project	Landside planning and management
Simulation Output	Capacity, flow rate, delay, bottleneck
Visualization Capabilities	Yes

Total Airspace and Airport Modeler (TAAM)

Scope	Airspace, runway, taxiway, gate (gate to gate)
Fidelity	High
Functionality and Capabilities	Airport capacity (gate, taxiway, runway capacity), deicing, noise impact, impact of severe weather, design of SIDs and STARs, design of ATC sectors, ATC workload.
Applicable Project	Airspace planning, airside planning
Simulation Output	System delays, conflicts, airport movements, operations on taxiways and runways, runway occupancy, noise contours, fuel burn, ATC workloads, individual AC profiles.
Visualization Capabilities	Yes

PTV Vissim

Scope	Roadway, curbside
Fidelity	High
Functionality and Capabilities	Microscopic simulation of car, train, and person. It is represented by a corresponding entity in the simulation, thereby considering all relevant properties.
Applicable Project	Landside planning and management
Simulation Output	Capacity, throughput, delay by each entity
Visualization Capabilities	Yes

Copyright National Academy of Sciences. All rights reserved.

Copyright National Academy of Sciences. All rights reserved.

Abbreviations and acronyms used without definitions in TRB publications:

A4A	Airlines for America
AAAE	American Association of Airport Executives
AASHO	American Association of State Highway Officials
AASHTO	American Association of State Highway and Transportation Officials
ACI–NA	Airports Council International–North America
ACRP	Airport Cooperative Research Program
ADA	Americans with Disabilities Act
APTA	American Public Transportation Association
ASCE	American Society of Civil Engineers
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
ATA	American Trucking Associations
CTAA	Community Transportation Association of America
CTBSSP	Commercial Truck and Bus Safety Synthesis Program
DHS	Department of Homeland Security
DOE	Department of Energy
EPA	Environmental Protection Agency
FAA	Federal Aviation Administration
FAST	Fixing America's Surface Transportation Act (2015)
FHWA	Federal Highway Administration
FMCSA	Federal Motor Carrier Safety Administration
FRA	Federal Railroad Administration
FTA	Federal Transit Administration
HMCRP	Hazardous Materials Cooperative Research Program
IEEE	Institute of Electrical and Electronics Engineers
ISTEA	Intermodal Surface Transportation Efficiency Act of 1991
ITE	Institute of Transportation Engineers
MAP-21	Moving Ahead for Progress in the 21st Century Act (2012)
NASA	National Aeronautics and Space Administration
NASAO	National Association of State Aviation Officials
NCFRP	National Cooperative Freight Research Program
NCHRP	National Cooperative Highway Research Program
NHTSA	National Highway Traffic Safety Administration
NTSB	National Transportation Safety Board
PHMSA	Pipeline and Hazardous Materials Safety Administration
RITA	Research and Innovative Technology Administration
SAE	Society of Automotive Engineers
SAFETEA-LU	Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (2005)
TCRP	Transit Cooperative Research Program
TDC	Transit Development Corporation
TEA-21	Transportation Equity Act for the 21st Century (1998)
TRB	Transportation Research Board
TSA	Transportation Security Administration
U.S. DOT	United States Department of Transportation

Copyright National Academy of Sciences. All rights reserved.

TRANSPORTATION RESEARCH BOARD
500 Fifth Street, NW
Washington, DC 20001

ADDRESS SERVICE REQUESTED

NON-PROFIT ORG.
U.S. POSTAGE
PAID
COLUMBIA, MD
PERMIT NO. 88

National Academy of Sciences. All rights reserved.

The National Academies of
SCIENCES • ENGINEERING • MEDICINE

The nation turns to the National Academies of Sciences, Engineering, and Medicine for independent, objective advice on issues that affect people's lives worldwide.
www.national-academies.org

ISBN 978-0-309-48066-6
90000
9 780309 480666